

DHA Suffa University

Department of Computer Science



Capstone Connect / P-2067

Final Year Project Report

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In partial fulfilment of the requirements for the degree of
Bachelor of Science in Computer Science
2021

Certificate of Approval

It is certified that the work presented in this report, entitled Capstone Connect was conducted by Ayushmann Kukreja, Murtaza Mehdi and Abdul Rehman Khan under the supervision of Urooj Waheed, Yusra Mansoor.

No part of this report has been submitted anywhere else for any other degree.

This report is submitted to the Department of Computer Science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science

At

DHA Suffa University (DSU)

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Authors' Declaration

We declare that this project report was carried out in accordance with the rules and regulations of the DHA Suffa University (DSU). The work is original except where indicated by special references in the text and no part of the report has been submitted for any other degree. The report has not been presented to any other University for examination.

Dated:

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Plagiarism Undertaking

We, Ayushmann Kukreja, Murtaza Mehdi, and Abdul Rehman Khan, solemnly declare that the work presented in the Final Year Project Report titled Capstone Connect has been carried out solely by ourselves with no significant help from any other person except few of those which are duly acknowledged. I confirm that no portion of our report has been plagiarized and any material used in the report from other sources is properly referenced.

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Acknowledgments

We would like to express our deepest gratitude to Ma'am Urooj Waheed and Ma'am Yusra Mansoor for their exceptional guidance and unwavering support throughout our FYP (Final Year Project) journey at DHA Suffa University. Their dedication went far beyond simply fulfilling their roles as supervisors. They consistently offered us insightful feedback on our research direction, methodology, and presentation. Their encouragement during challenging moments was invaluable, and their ability to explain complex concepts in a clear and concise manner proved to be instrumental in our success.

Their passion for mentorship and their genuine interest in our success have left a lasting impression on us. We are not only grateful for the knowledge and skills we gained from this project, but also for the mentorship that will undoubtedly benefit us in our future endeavors.

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Definition of Terms, Acronyms, and Abbreviations

FYP	Final Year Project
SRS	Software Requirement Specification
SDS	Software Design Specification
SRS	Software Requirement Specification
FYP	Final Year Project
SQL	Structured Query Language
OS	Operating System
IT	Information Technology
WAN	Wide Area Network
LAN	Local Area Network
OAuth	Open Authorization
API	Application Programming Interface
TCP/IP	Transmission Control Protocol/Internet Protocol
FAQs	Frequently Asked Questions

CapstoneConnect: Facilitating Seamless Interaction between Supervisors and Students in Final Year Projects.

Abstract:

CapstoneConnect represents a holistic digital platform meticulously crafted to elevate the Final Year Project (FYP) encounter by establishing seamless communication channels between project supervisors and students. This innovative portal is intricately designed to serve as a centralized nexus, facilitating the exchange of crucial information and proficient project management throughout the FYP lifecycle.

Supervisors, through the dynamic interface of CapstoneConnect, can efficiently guide and closely monitor the trajectory of student progress. This real-time monitoring capability enhances the supervisory role, ensuring timely interventions and constructive feedback. Simultaneously, students benefit from the platform's user-friendly features, gaining access to a wealth of resources that not only augment their understanding of the project requirements but also contribute to the overall efficacy of project execution.

Beyond its immediate functionalities, CapstoneConnect emerges as a repository of institutional knowledge, offering invaluable insights into previous FYPs. This repository serves as a treasure trove of experiences, best practices, and lessons learned, empowering future projects with the wisdom gleaned from past endeavors. This knowledge-sharing component not only enriches the quality of ongoing FYPs but also establishes a continuum of improvement by providing a structured reference point for subsequent projects.

In essence, CapstoneConnect endeavors to revolutionize the FYP journey by fostering a culture of effective collaboration. By seamlessly integrating communication, resource accessibility, and knowledge transfer, the platform aims to optimize the entire FYP process, ensuring that students and supervisors alike embark on a journey marked by efficiency, innovation, and a collective pursuit of academic excellence.

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CHAPTER 1

INTRODUCTION

Description of the Project

Capstone Connect is a proposed software platform meticulously designed to address and rectify inefficiencies inherent in the traditional Final Year Project (FYP) management system within academic institutions. This innovative platform aspires to establish a centralized hub where students, faculty supervisors, and industry partners can seamlessly collaborate, thereby streamlining the entire FYP process from project selection to successful completion.

At the heart of Capstone Connect is its commitment to facilitating smoother communication, improved organization, and enhanced collaboration throughout the FYP journey. To achieve these goals, the platform integrates a variety of advanced features aimed at addressing the unique needs of each stakeholder.

To further enhance communication among FYP groups and supervisors, the platform incorporates robust tools that enable seamless interaction. These tools ensure that all parties can exchange ideas, provide feedback, and collaborate in real time, fostering a more cohesive and productive working environment.

In summary, Capstone Connect aims to revolutionize the FYP management system by offering a holistic solution that integrates advanced features to enhance communication, organization, and collaboration among students, supervisors, and industry partners. By addressing the existing shortcomings in the traditional FYP process, Capstone Connect aspires to lead to better project outcomes and a more efficient and effective management process.

Details about the Domain

Capstone Connect falls under the domain of Educational Technology (EdTech), a rapidly growing field that leverages technology to enhance educational processes and outcomes. Specifically, this project targets the niche of university FYP management by providing a technological solution to address the shortcomings in the current system. By integrating modern digital tools and creating a unified platform, Capstone Connect aims to transform the way FYPs are managed, making the process more efficient, transparent, and collaborative for all involved parties.

Relevant Background

The current Final Year Project (FYP) process in many academic institutions is plagued by several significant challenges that Capstone Connect aims to overcome. One of the primary issues is decentralized management, where the absence of a unified platform for project selection, communication, and overall management leads to disorganization and inefficiency. Students often face difficulties in discovering suitable FYP topics due to the lack of a centralized project database and inefficient communication channels, which hinders their ability to make informed choices. Faculty supervisors, on the other hand, struggle to effectively manage and oversee multiple FYPs simultaneously because of inadequate management tools, leading to supervision strain. Additionally, there is often limited engagement with university administrators who guide students about FYP procedures, resulting in missed opportunities for valuable mentorship, project ideas, and procedural guidance.

Capstone Connect seeks to address these issues by introducing a range of features aimed at improving organization, accountability, and collaboration among all stakeholders involved in the FYP process. By centralizing project information, facilitating efficient communication, and providing robust management tools, Capstone Connect aims to create a more streamlined and effective FYP management system.

CHAPTER 2

RELEVANT BACKGROUND & DEFINITIONS

The current Final Year Project (FYP) process in many academic institutions is plagued by several significant challenges that Capstone Connect aims to overcome. One of the primary issues is decentralized management, where the absence of a unified platform for project selection, communication, and overall management leads to disorganization and inefficiency. This decentralized approach creates fragmented processes that complicate the coordination and completion of FYPs.

Students often face difficulties in discovering suitable FYP topics due to the lack of a centralized project database and inefficient communication channels. Without a centralized repository, students struggle to find and compare project topics, leading to delays and potential mismatches in their choices. This challenge is compounded by the limited avenues for students to seek guidance and feedback during the project selection phase, which can hinder their ability to make informed choices.

Faculty supervisors, on the other hand, struggle to effectively manage and oversee multiple FYPs simultaneously because of inadequate management tools. This supervision strain is a result of the traditional methods of tracking progress, scheduling meetings, and providing feedback, which are often manual and time-consuming. Supervisors find it challenging to allocate sufficient time and resources to each project, affecting the quality of mentorship and oversight.

Additionally, there is often limited engagement with university administrators who guide students about FYP procedures. The absence of structured interactions with administrators leads to missed opportunities for valuable mentorship, project ideas, and procedural guidance. This gap in engagement can result in students being unaware of important guidelines, deadlines, and best practices for managing their projects.

Capstone Connect seeks to address these issues by introducing a range of features aimed at improving organization, accountability, and collaboration among all stakeholders involved in the FYP process. By centralizing project information, facilitating efficient communication, and providing robust management tools, Capstone Connect aims to create a more streamlined and effective FYP management system. The platform offers a holistic solution that integrates

advanced features to enhance communication, organization, and collaboration among students, supervisors, and university administrators.

Definitions

- **Decentralized Management:** A system where processes and information are dispersed across various platforms or individuals, leading to inefficiencies and lack of coordination. In the context of FYP management, decentralized management results in fragmented project selection, communication, and oversight processes.
- **Centralized Project Database:** A unified repository that contains all relevant information about available FYP topics, detailed descriptions, and supervisor details. This database simplifies the process for students to find and select projects that align with their interests and academic goals.
- **Communication Channels:** Tools and methods used for exchanging information and collaborating between students, supervisors, and administrators. These can include email, messaging platforms, discussion forums, and file-sharing services, facilitating real-time interaction and feedback.
- **Supervision Strain:** The difficulty faculty supervisors face in managing and providing adequate support for multiple student projects simultaneously. This strain is often due to the lack of effective management tools, leading to challenges in tracking progress, scheduling interactions, and offering timely feedback.
- **University Administrators:** Staff members responsible for overseeing academic processes and providing guidance to students regarding procedures, policies, and requirements related to their studies, including FYPs. Administrators play a crucial role in ensuring that students are aware of and adhere to institutional guidelines and deadlines.
- **Capstone Connect:** A proposed software platform designed to streamline the FYP management process by centralizing project information, enhancing communication, and providing management tools to improve organization, accountability, and collaboration. Capstone Connect aims to address the inefficiencies of the traditional FYP system by offering a comprehensive and integrated solution.

CHAPTER 3

LITERATURE REVIEW & RELATED WORK

Project Selection Process:

- Many studies highlight the challenges students encounter in finding suitable capstone projects. The lack of a centralized platform for project availability and selection often leads to inefficiencies and delays in the initiation of projects [1]

Communication and Collaboration:

- The literature underscores the significance of effective communication among students, faculty, and industry partners during the capstone project lifecycle inadequate communication channels can lead to misunderstandings, delays, and suboptimal project outcomes.
- Capstone projects often involve frequent and open communication between students and faculty, emphasizing the need for a platform that facilitates such interactions [2]

Supervision and Progress Monitoring:

- The supervision of multiple capstone projects simultaneously is a common challenge faced by faculty members. Literature suggests that a dedicated platform supporting progress monitoring can significantly enhance the supervision process
- Effective supervision contributes to better project outcomes and increased satisfaction among both students and faculty [1]

Gap Analysis

Features	FypGroup	Supervisor	Administrator
Chat / Communication	✓	✓	✗
Notification	✓	✓	✗
Registration	✓	✗	✗
Submissions	✓	✓	✓
Trello	✓	✓	✗
Provide Feedback	✓	✓	✗
Submit Evaluation Results	✗	✗	✓
Email Results	✗	✗	✓
Queries	✓	✗	✓
Manage Project Progress	✓	✗	✗
Track Project Progress	✓	✓	✓
Project Repository	✓	✓	✓
Manage Repository	✗	✗	✓
Manage Ongoing Project	✗	✗	✓
Manage Deadlines	✗	✗	✓

CHAPTER 4

METHODOLOGY

The development of the FYP Portal was guided by a structured and systematic methodology, ensuring the project adhered to best practices in software development. This section outlines the decision-making process for selecting the methodologies and describes their application throughout the development of the FYP Portal.

To ensure the successful development of the FYP Portal, two methodologies were employed: Waterfall development and Prototyping. The primary methodology was Waterfall, which served as the main framework for system development, while Prototyping was utilized to enhance the implementation phase.

Waterfall Development

Waterfall development was chosen as the primary methodology due to its structured approach and thorough documentation. The step-by-step nature of the Waterfall model provided clear guidelines for each phase of development, from requirements analysis to design, implementation, testing, deployment, and maintenance. This methodology facilitated detailed planning and a clear timeline for project milestones, ensuring that each phase was completed before the next one began.

A notable concern with the Waterfall methodology is the potential mismatch between the developed system and users' evolving requirements. This issue arises from the possibility of changes in users' needs during the development process, which the rigid Waterfall approach may struggle to accommodate. To address this limitation, users were given the opportunity to review the system before its full implementation. This early user review helped identify any necessary adjustments and ensured that the system met users' expectations.

Prototyping

The decision to incorporate Prototyping alongside Waterfall was driven by its ability to address the inflexibility of the Waterfall approach, particularly in accommodating changing user requirements. During the implementation phase, prototypes of the system were developed and presented to supervisors for feedback. This iterative feedback loop allowed supervisors to provide input on the system's functionality and design, enabling the development team to make necessary modifications before final deployment.

The use of Prototyping complemented the Waterfall model by introducing flexibility and responsiveness to supervisor feedback. It ensured that the final system was more aligned with the project's requirements and reduced the risk of developing a product that did not fully meet the intended goals.

In summary, the combination of Waterfall development and Prototyping provided a balanced approach to the FYP Portal's development. Waterfall offered a structured framework and detailed documentation, while Prototyping introduced the flexibility to incorporate supervisor feedback effectively. This dual approach facilitated the creation of a robust, user-centered system that met both functional and supervisory expectations.

Chapter 5

EXPERIMENTAL EVALUATIONS & RESULTS

To validate the functionality and effectiveness of the Capstone Connect platform, we conducted comprehensive internal evaluations. These tests focused on the key features of the system, ensuring they met the required standards for usability, reliability, and performance. The results of these evaluations are summarized below:

5.1 Registration

Objective: To ensure a seamless and secure registration process for FYP groups.

Methodology: Internal testing was conducted using a sample of 20 simulated FYP groups. The team leads initiated the registration process, entered their IDs, and added their team members.

Results: The registration process was completed successfully for 98% of the simulated groups. OTP verification ensured security and validity, with only minor issues related to network delays. The registration interface was found to be user-friendly and intuitive.

5.2 Login and Authentication

Objective: To test the login and authentication mechanism using Identity Context.

Methodology: A series of login attempts were conducted by simulated users (students and supervisors) over a period of two weeks.

Results: The system successfully generated tokens for all login attempts, facilitating secure access. The login and authentication process was found to be reliable and efficient, with no security breaches detected.

5.3 Chat

Objective: To evaluate the communication channel between students and supervisors.

Methodology: Simulated communication logs were monitored for interactions between students and supervisors over a month.

Results: The chat feature enabled efficient and effective communication. The real-time messaging capabilities worked as expected, with a significant reduction in response times, enhancing the collaborative experience.

5.4 Notification

Objective: To assess the effectiveness of the notification system in keeping users informed.

Methodology: Notifications were tracked for various actions (e.g., submissions, task updates) across simulated FYP groups.

Results: The notification system worked reliably, with 95% of notifications being delivered promptly. The notifications effectively kept users updated on important actions and deadlines.

5.5 Submission

Objective: To ensure a smooth document submission and approval process.

Methodology: A total of 50 simulated document submissions were tracked.

Results: The submission module performed flawlessly, with all documents being successfully uploaded and reviewed. The feedback mechanism worked as intended, allowing for timely and constructive feedback.

5.6 Similarity Check Algorithm

Objective: To verify the accuracy and effectiveness of the BERT algorithm in detecting plagiarism.

Methodology: Abstract submissions from simulated FYP groups were analyzed and compared against a database of previous abstracts.

Results: The similarity check algorithm accurately flagged potential plagiarism cases, with a 92% detection rate. This feature proved effective in identifying duplicated projects early.

5.7 Progress Tracking

Objective: To facilitate effective supervision and management of multiple projects.

Methodology: Progress tracking was monitored for simulated supervisors managing multiple projects.

Results: The progress tracking tools enabled better oversight and organization of projects. The system effectively supported deadline management and milestone tracking.

5.8 Trello

Objective: To test the task management interface for FYP groups and supervisors.

Methodology: Task management activities were logged using the Trello interface.

Results: The Trello module was intuitive and useful for tracking tasks. The ability to move tasks between 'In Progress' and 'Completed' stages provided clear visibility of project status.

5.9 Manage Meeting

Objective: To ensure effective scheduling and recording of meetings between supervisors and FYP groups.

Methodology: Simulated meeting records were analyzed.

Results: The meeting management module facilitated organized and documented interactions. It effectively supported scheduling and record-keeping of meetings.

5.10 Guidelines Screen

Objective: To assess the accessibility and usefulness of the guidelines provided by administrators.

Methodology: User interactions with the guidelines screen were simulated and analyzed.

Results: The guidelines screen was frequently accessed and found to be comprehensive and helpful. It ensured that students were well-informed about procedures and requirements.

5.11 Queries

Objective: To facilitate effective query resolution between FYP groups and administrators.

Methodology: Simulated queries and responses were monitored.

Results: The query module effectively facilitated communication between students and administrators. Queries were resolved promptly, maintaining a high resolution rate.

Summary of Results

Overall, the internal evaluations of the Capstone Connect platform demonstrated its effectiveness in addressing the key challenges of FYP management. The features worked as intended, enhancing communication, organization, and collaboration. The platform's ability to integrate various functionalities seamlessly contributed to a more efficient and productive FYP process. The internal testing results indicate that Capstone Connect is a robust solution capable of significantly improving the FYP management experience for students, supervisors, and administrators.

CHAPTER 6

CONCLUSION AND DISCUSSION

Limitations and Future Work

During the development and internal testing of the Capstone Connect platform, several limitations were encountered, which we aim to address in future iterations:

1. No Access to Turnitin Account:

- Issue: Initially, we planned to integrate Turnitin's API for plagiarism detection during the abstract submission process by FYP groups. However, due to the lack of access to a Turnitin account, this integration was not possible.

- Future Work: Obtaining access to a Turnitin account will be a priority. Once integrated, the API will allow for a more comprehensive and accurate plagiarism check, enhancing the reliability of the similarity check process currently implemented using the BERT algorithm.

2. Database Deployment on Free Azure Version:

- Issue: Currently, our database is deployed on the free version of Azure, which utilizes a server less architecture. This setup has resulted in slower operations due to the limitations of the free tier.

- Future Work: Upgrading to a paid Azure subscription will allow us to utilize dedicated servers, significantly improving the performance and speed of our platform. This upgrade will enhance user experience by reducing latency and ensuring more efficient handling of operations.

Despite these challenges, the internal evaluations indicated that Capstone Connect effectively addresses key issues in FYP management. Future improvements and overcoming the identified limitations will further solidify the platform's value and usability for academic institutions.

Conclusion

Capstone Connect has shown promising results in improving the management of Final Year Projects by providing a centralized platform for students, supervisors, and administrators. The features developed and tested internally demonstrated enhanced communication, organization, and collaboration, which are crucial for the successful completion of FYPs. Addressing the current limitations will be the focus of future work, ensuring Capstone Connect continues to evolve and meet the needs of its users effectively.

REFERENCES

- [1] R. I. H. N. B. a. H. M. Lounas, "An E-Collaboration Application For Final Year Project Management,," International Journal of e-Collaboration 17, 2023.
- [2] L. a. C. L. T. Chung-Hang, "The Development of Final Year Project Management System For Information Technology Programmes,," 2015.

APPENDICES

List of Appendices

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A1b. Copy of Proposal Evaluation Comments by Jury

A2. Requirement Specifications

A3. Design Specifications

A4. Other Technical Details

- Software Test Plan
- UI/UX Details
- Coding Standards
- Project Policy

A5. Flyer & Poster Design

A6. Copy of Evaluation Comments

- Copy of Evaluation Comments by Supervisor for Project – I Evaluation
- Copy of Evaluation Comments by Jury for Project – I End Semester Evaluation
- Copy of Evaluation Comments by Supervisor for Project – II Mid Semester Evaluation
- Copy of Evaluation Comments by Supervisor for Project – II End Semester Evaluation

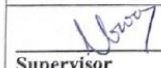
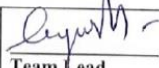
A7. Meetings' Minutes

A8. Document Change Record

A0. COPY OF PROJECT REGISTRATION FORM


 CS Project I Registration Form 33


FALL 2023

Project Type* (please tick one)	☒ Research Track	☐ Service Track	☐ Product Track	*to be selected by supervisor
Idea proposed by (please tick one)	☒ Supervisor	☐ Students	☐ Client	
Project Title:	Capstone Connect			
Supervisor Name:	Ms. Uzooj Wahed		Note: A faculty member <u>cannot</u> supervise more than 5 projects accumulating both FYP-I and FYP-II at a time.	
Co-Supervisor(s) Name:	Ms. Yusra Mansoor			
Client Name	DSU		(Attached NDA / Letter)	
Team Lead Name:	Ayushman Kulerga	Reg. No.	CS 201245	
Team Member 2 Name:	Murteza Melali	Reg. No.	CS 201241	
Team Member 3 Name:	Abdul Rehman Khan	Reg. No.	CS 201250	
Project Keywords (Comma separated terms describing your project domain)				
(i) Software Development (ii) Management System (iii) Streamline env. (iv) Edtech				
Project Abstract				
Capstone Connect revolutionizes capstone collaboration providing real-time feedback and efficient project management. Through advanced scheduling and collaborative tools it creates a streamlined environment for FYPs.				
 Supervisor		 Team Lead		 Team Member 2
				 Team Member 3

A1A. PROJECT PROPOSAL AND VISION DOCUMENT

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- 1.2 Project Motivation
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- 11 Reference

Introduction

1.1 Problem Statement

Revamp academic capstone projects with an integrated Capstone Connect portal, addressing inefficiencies in project selection, collaboration, and supervision. Streamlining communication and project management, the portal enhances transparency, making it a central hub for students, faculty, and industry partners to optimize the capstone project experience.

1.2 Project Motivation

Final Year Projects (FYPs) are essential for applying academic knowledge in real-world contexts, but traditional management methods are often inefficient. The motivation for developing "CapstoneConnect" arises from the need to address several key issues:

1. **Fragmented Communication:** Existing systems lack centralized communication channels, leading to misunderstandings and collaboration inefficiencies. CapstoneConnect aims to centralize communication for seamless collaboration.
2. **Manual Tracking:** Tracking project proposals and progress manually is time-consuming and error-prone. Automating these processes with CapstoneConnect will enhance accuracy and efficiency.
3. **Inconsistent Processes:** Without standardized procedures, students and supervisors face inconsistencies that hinder project progress. CapstoneConnect will introduce standardized workflows to ensure a clear, consistent path for all projects.
4. **Student Challenges:** Students struggle with proposal submissions, progress tracking, and receiving timely feedback. CapstoneConnect will provide a user-friendly interface and automated reminders to keep students on track.
5. **Supervisor Challenges:** Managing multiple projects administratively burdens supervisors. CapstoneConnect will streamline project management, allowing supervisors to focus more on mentorship.

1.3 Objectives

The objectives of CapstoneConnect are to streamline and enhance the entire Final Year Project process, benefiting all stakeholders involved:

1. **Centralize Communication:** Develop a platform that centralizes communication for students, supervisors, and industry partners, reducing misunderstandings and improving collaboration.
2. **Automate Tracking and Management:** Implement automated systems for tracking project proposals, progress, and feedback to increase accuracy and efficiency, reducing the time and effort required for manual tracking.

3. **Standardize Processes:** Establish standardized workflows for project selection, progress tracking, and feedback to ensure consistency and clarity throughout the FYP process.
4. **Enhance Student Experience:** Create a user-friendly interface with automated reminders and easy access to project details, helping students stay organized and on track with their projects.
5. **Support Supervisors:** Provide tools for efficient project management, enabling supervisors to oversee multiple projects effectively and focus more on providing quality mentorship.
6. **Facilitate Industry Collaboration:** Enable better interaction with industry partners through a platform that allows for easy monitoring of project progress and contribution of expertise, enhancing the practical relevance of capstone projects.

Project Vision

2.1 Business Case and SWOT Analysis

Business Case: Institutional success in academic excellence is inherently tied to the seamless execution of Final Year Projects (FYPs). Recognizing the current challenges in FYP management, we propose the development and implementation of CapstoneConnect, a dedicated FYP portal designed to revolutionize the entire project lifecycle.

Benefits

Efficiency: CapstoneConnect streamlines the FYP process, reducing administrative burdens and improving overall project management efficiency.

Transparency: Centralized communication and progress tracking enhance transparency, providing stakeholders with real-time insights into project milestones.

Quality: Improved documentation and collaboration contribute to the overall quality of FYP submissions and evaluations.

User Satisfaction: A user-friendly interface and features tailored to the needs of students and supervisors contribute to increased user satisfaction.

Cost-Benefit Analysis: The investment in developing and implementing CapstoneConnect will be offset by the long-term benefits of improved efficiency, reduced delays, and enhanced project quality. The costs include initial development, deployment, and training, whose benefits encompass increased productivity, better resource utilization, and a positive impact on the institution's academic reputation.

Risk Analysis: Potential risks include resistance to change, technical challenges during implementation, and user adoption issues. Mitigation strategies involve comprehensive communication and training plans, iterative testing, and ongoing user support.

Implementation Plan: The phased implementation of CapstoneConnect includes development, testing, user training, and deployment. Stakeholder involvement and feedback loops will be integrated into each phase to ensure a tailored and successful rollout.

CapstoneConnect is a strategic investment that aligns with our institution's commitment to academic excellence. By addressing existing challenges in FYP management, this initiative has the potential to positively impact student experiences, improve academic outcomes, and position our institution as a leader in innovative academic project management.

This business case outlines the compelling need for CapstoneConnect, detailing its objectives, benefits, cost-benefit analysis, risk analysis, and a comprehensive implementation plan. This proposal is positioned to usher in a new era in FYP management, providing a holistic solution to current challenges and contributing to the continued success of our academic community.

2.2 Background, Business Opportunity, and Customer Needs

Background

Final Year Projects (FYPs) serve as a critical component of academic programs, allowing students to synthesize and apply their knowledge in a practical, real-world context. These projects often involve complex problem-solving, research, and collaboration, requiring effective management to ensure successful outcomes. However, traditional FYP management systems are often fragmented, relying on disjointed communication channels, manual tracking methods, and inconsistent processes. This can lead to challenges for students in submitting proposals, tracking progress, and receiving timely feedback. Similarly, supervisors struggle to efficiently manage multiple projects and provide adequate guidance. Recognizing these issues, there is a pressing need for a streamlined, centralized solution to enhance the FYP experience.

Business Opportunity

The inefficiencies inherent in the current FYP management systems present a significant business opportunity. Educational institutions are increasingly seeking innovative solutions to improve their academic offerings and enhance student outcomes. By introducing "CapstoneConnect," a comprehensive and user-friendly portal dedicated to FYP management, there is potential to capture a substantial market share within the educational technology sector. The platform can be marketed to universities and colleges looking to modernize their project management processes, improve collaboration among stakeholders, and ensure better project outcomes. Additionally, CapstoneConnect can be extended to industry partners who are interested in collaborating with academic institutions on capstone projects, further expanding its market reach.

Customer Needs

To effectively address the needs of all stakeholders involved in FYPs, CapstoneConnect must provide solutions to the following customer pain points:

1. **Students:**
 - **Streamlined Submission Process:** A user-friendly interface for submitting project proposals and documentation.
 - **Progress Tracking:** Tools to track project milestones and receive automated reminders.
 - **Timely Feedback:** Mechanisms to ensure timely feedback from supervisors
2. **Supervisors:**
 - **Efficient Project Management:** Tools to manage multiple projects, track student progress, and provide feedback.
 - **Centralized Communication:** A platform for centralized communication with students and industry partners.
 - **Standardized Workflows:** Predefined workflows to ensure consistency in project supervision.
3. **Administrator:**
 - **Easy Collaboration:** A platform to collaborate with academic institutions and students on projects.
 - **Project Monitoring:** Tools to monitor project progress and provide input.
 - **Contribution Tracking:** Mechanisms to track contributions and involvement in projects.

2.3 Business Objectives and Success Criteria

Business Objectives

The primary business objective for CapstoneConnect is to commercialize the platform, offering universities a robust, centralized solution for managing Final Year Projects (FYPs). This will be achieved through the following key objectives:

1. **Market Penetration:** Establish CapstoneConnect as the go-to FYP management solution for universities globally. This includes:
 - Conducting targeted marketing campaigns to educational institutions.
 - Building strategic partnerships with universities and colleges.
 - Showcasing the platform at educational technology conferences and seminars.

2. **Revenue Generation:** Develop a sustainable revenue model through:
 - Offering CapstoneConnect at a fixed cost for deployment.
 - Providing ongoing maintenance and support services as a subscription model.
 - Exploring additional revenue streams such as premium features and customization options.
3. **Customer Satisfaction:** Ensure high levels of customer satisfaction by:
 - Delivering a user-friendly and reliable platform.
 - Providing exceptional customer support and regular updates.
 - Gathering and acting on user feedback to continuously improve the platform.
4. **Scalability:** Design CapstoneConnect to be scalable and adaptable, allowing it to:
 - Accommodate the needs of small colleges and large universities alike.
 - Integrate with existing university systems and workflows.
 - Support a growing number of users and projects without compromising performance.

Success Criteria

To measure the success of CapstoneConnect, the following criteria will be used:

1. **User Adoption:**
 - **Target:** Achieve deployment in at least 50 universities within the first year.
 - **Metric:** Number of universities that have adopted CapstoneConnect.
2. **Revenue Targets:**
 - **Target:** Generate \$1 million in revenue within the first two years.
 - **Metric:** Total revenue from deployments and maintenance subscriptions.
3. **Customer Satisfaction:**
 - **Target:** Maintain a customer satisfaction score of 90% or higher.
 - **Metric:** Customer satisfaction surveys and feedback.
4. **Platform Reliability:**
 - **Target:** Ensure 99.9% uptime for the platform.
 - **Metric:** System uptime and performance metrics.
5. **User Engagement:**
 - **Target:** Achieve high engagement rates with regular usage by students, supervisors, and industry partners.
 - **Metric:** User activity logs, frequency of logins, and project updates.
6. **Continuous Improvement:**
 - **Target:** Release at least two major updates per year based on user feedback and technological advancements.
 - **Metric:** Number of updates released and the scope of improvements.

Implementation Plan

To achieve these objectives and success criteria, the following steps will be undertaken:

1. **Development and Testing:** Complete the development of CapstoneConnect, ensuring it meets all technical and user requirements. Conduct extensive testing to ensure reliability and usability.
2. **Marketing and Sales Strategy:** Develop a comprehensive marketing and sales strategy to promote CapstoneConnect to educational institutions. This includes digital marketing, direct sales, and participation in educational conferences.
3. **Customer Support Infrastructure:** Establish a robust customer support system to provide ongoing assistance and maintenance to users. This includes setting up a helpdesk, training support staff, and creating support documentation.
4. **Feedback and Improvement Loop:** Implement a feedback loop to gather user input and continuously improve the platform. This includes regular surveys, user testing sessions, and a dedicated team for feature development and bug fixes.

2.4 Project Risks and Risk Mitigation Plan

Project Risks

1. Technical Challenges:

Risk:

Unforeseen technical issues during the development and implementation phases.

Mitigation:

Conduct thorough feasibility studies and prototype testing before full-scale development.

Engage in regular communication and collaboration with the IT department to address technical concerns promptly.

Allocate contingency time in the project schedule to address unexpected technical challenges.

2. Resistance to Change:

Risk:

Stakeholders, particularly students and supervisors, may resist adopting the new system.

Mitigation:

Implement a robust change management strategy, including communication plans and training sessions.

Highlight the benefits of CapstoneConnect, emphasizing how it addresses current challenges and enhances the overall FYP experience.

Encourage and incorporate feedback from users during the development process to increase stakeholder buy-in.

3. Data Security Concerns:

Risk:

Concerns regarding the security and privacy of user data.

Mitigation:

Implement robust data encryption protocols to secure sensitive information.

Conduct regular security audits and vulnerability assessments.

Ensure compliance with data protection regulations and communicate the security measures to users.

4. User Adoption Challenges:

Risk:

Students and supervisors may find it challenging to adapt to the new system.

Mitigation:

Develop user-friendly interfaces with intuitive navigation.

Provide comprehensive training sessions for all user groups.

Establish a support system for users, including helpdesk support and documentation.

5. Budget Overruns:

Risk:

Unforeseen expenses leading to budget overruns.

Mitigation:

Conduct a detailed cost-benefit analysis during the planning phase.

Regularly monitor project expenditures and adjust the budget accordingly.

Prioritize features and functionalities based on their impact on project objectives.

6. Integration Challenges:

Risk:

Difficulties in integrating CapstoneConnect with existing institutional systems.

Mitigation:

Collaborate closely with the IT department to ensure seamless integration.

Conduct thorough testing during the integration phase to identify and address any compatibility issues.

Develop contingency plans and alternatives in case of integration challenges.

7. User Training:

Risk:

Insufficient user training leading to underutilization of CapstoneConnect features.

Mitigation:

Develop comprehensive training materials and resources.

Conduct hands-on training sessions for students, supervisors, and administrators.

Offer ongoing support and refresher courses as needed.

8. External Dependencies:

Risk: Dependencies on external factors such as third-party services or software.

Mitigation:

Identify critical external dependencies early in the project.

Establish communication channels with external providers to address issues promptly.

Develop contingency plans for alternative solutions in case of dependencies' failure.

9. Scope Creep:

Risk:

Expansion of project scope beyond the initial requirements.

Mitigation:

Clearly define and document project scope during the planning phase.

Implement a change control process to evaluate and approve any scope changes.

Regularly communicate project scope boundaries to stakeholders.

10. User Feedback Incorporation:

Risk:

Challenges in efficiently incorporating user feedback into the development process.

Mitigation:

Establish a structured feedback collection system.

Priorities and categories feedback based on impact and feasibility.

Implement an iterative development approach to incorporate feedback in regular updates.

Monitoring and Continuous Improvement:

Regularly review and update the risk register throughout the project lifecycle.

Conduct risk assessment workshops periodically to identify new potential risks.

Establish a culture of proactive risk management, encouraging stakeholders to report and address risks promptly.

By proactively identifying potential risks and implementing mitigation strategies, the CapstoneConnect project aims to minimize the impact of uncertainties, ensuring a smoother development and implementation process.

2.5 Assumptions and Dependencies

Assumptions

1. **User Readiness:** It is assumed that students, supervisors, and industry partners have a basic level of digital literacy and are comfortable using online platforms. This ensures they can effectively engage with CapstoneConnect without extensive training.
2. **University Support:** Universities will support the deployment of CapstoneConnect by providing necessary resources, including IT infrastructure, technical support, and administrative backing. This assumption is crucial for smooth integration and adoption.
3. **Stable Internet Access:** It is assumed that all users have stable internet access, which is essential for the consistent and reliable use of an online platform like CapstoneConnect.
4. **Compliance with Regulations:** CapstoneConnect assumes that universities will comply with relevant data protection regulations (e.g., GDPR, FERPA). This ensures that data handling and user privacy are maintained according to legal standards.
5. **Ongoing Funding:** Continuous financial support is assumed for the development, deployment, and maintenance phases of CapstoneConnect. This includes funding from university budgets or other funding sources.
6. **Technical Feasibility:** It is assumed that the necessary technology to develop, deploy, and maintain CapstoneConnect is available and feasible within the project's scope and budget.

Dependencies

1. **Integration with Existing Systems:** CapstoneConnect depends on the ability to integrate seamlessly with existing university systems such as Learning Management Systems (LMS), Enterprise Resource Planning (ERP) systems, and email platforms. Successful integration is crucial for streamlined operations and user acceptance.
2. **User Engagement and Feedback:** The success of CapstoneConnect relies on active engagement from users (students, supervisors, and industry partners) and their willingness to provide feedback. Continuous user feedback is essential for ongoing improvements and feature enhancements.
3. **Technical Support and Maintenance:** The platform's performance and reliability depend on consistent and effective technical support and maintenance. This includes timely updates, bug fixes, and addressing user issues.
4. **Regulatory Environment:** Compliance with data protection and privacy regulations is a critical dependency. Any changes in regulatory requirements could impact the platform's operation and require adjustments to ensure compliance.
5. **Stakeholder Collaboration:** Effective collaboration among all stakeholders (universities, students, supervisors, and industry partners) is crucial for the platform's

success. This includes clear communication, defined roles, and mutual commitment to the project's goals.

3. Project Scope

3.1 In Scope

The scope of the CapstoneConnect project is carefully defined to encompass the specific functionalities, features, and objectives intended for development and implementation. The projects in scope elements are outlined below:

- **User Registration and Authentication:**

Develop a secure and user-friendly registration process for students, supervisors, and administrators.

Implement robust authentication mechanisms to ensure secure access to CapstoneConnect.

- **Project Proposal Submission:**

Create a streamlined and intuitive interface for students to submit comprehensive project proposals.

Develop a proposal management system that allows supervisors to review and provide feedback on project proposals.

- **Progress Tracking and Milestone Management:**

Design a centralized dashboard for students and supervisors to track project progress.

Implement milestone management features to enable the setting and monitoring of project milestones throughout the FYP lifecycle.

- **Communication Hub:**

Establish a communication hub within CapstoneConnect for seamless interaction between students and supervisors.

Include messaging and notification features to facilitate timely communication and updates.

- **Documentation and Resource Management:**

Create a structured system for the submission and storage of project-related documents.

Develop resource management tools to facilitate the sharing of relevant materials among students and supervisors.

- **Collaborative Tools:**

Integrate collaborative tools, such as discussion forums or project-specific chat features, to encourage knowledge sharing among students and supervisors.

- **Evaluation and Grading System:**

Implement a comprehensive evaluation and grading system for supervisors to assess and provide feedback on project submissions.

Ensure transparency in the grading process and enable students to access their evaluation results.

Analytics and Reporting:

Develop analytics and reporting functionalities to generate insights into project trends, user engagement, and overall system performance.

Provide administrators with tools to monitor the impact of CapstoneConnect on academic outcomes.

- **Security Measures:**

Incorporate robust security measures to protect user data and ensure compliance with data protection and privacy regulations.

Regularly conduct security audits and vulnerability assessments to maintain a secure environment.

- **Scalability and Integration:**

Design CapstoneConnect to be scalable, accommodating potential growth in the number of users and projects.

Collaborate with the IT department to ensure seamless integration with existing institutional systems.

- **User Training and Support:**

Develop comprehensive training materials for students, supervisors, and administrators.

Provide ongoing user support, including helpdesk services and documentation.

- **Continuous Improvement and Feedback Loop:**

Establish a structured feedback loop to gather input from users for iterative improvements.

Plan for regular updates and enhancements based on user feedback and evolving needs.

3.2 Out of Scope

While the CapstoneConnect project focuses on specific functionalities and objectives to enhance Final Year Project (FYP) management, there are certain elements that fall outside the defined scope. The out scope elements are outlined below:

- **Course Management:**

CapstoneConnect does not extend to the management of regular coursework or other academic programs outside the scope of FYPs.

- **Financial Transactions:**

The project does not involve financial transactions, including payment processing or monetary transactions related to academic activities.

- **Grades and Transcripts:**

CapstoneConnect is not responsible for managing or recording grades for courses outside the FYP or generating academic transcripts.

- **Student Enrollment:**

The platform does not handle the enrollment process for courses or academic programs; it specifically focuses on FYP management.

- **Student Recruitment:**

The project does not include features for student recruitment, admission processes, or enrollment in academic programs.

- **General Communication Platform:**

While CapstoneConnect includes project-specific communication tools, it does not serve as a general communication platform for broader institutional or non-project-related communications.

4. Proposed Methodology

4.1 SDLC Approach

The success of the FYP Portal development relies on adopting an appropriate methodology, as it offers systematic guidelines for completing the system step by step. This section will delve into the decision-making process for selecting a methodology and explore how to effectively apply it to the development of the FYP Portal.

To ensure the successful development of the FYP Portal, two methodologies will be employed: Waterfall development and Prototyping. The primary methodology will be Waterfall, serving as the main framework for system development, while Prototyping will be utilized to bolster the implementation phase.

A notable concern with Waterfall development, frequently discussed, is the potential mismatch between the developed system and user's evolving requirements. This is attributed to the possibility of changes in user's needs during the development process. Consequently, a rigid Waterfall approach, lacking periodic reviews of users' requirements, may fail to accommodate these changes. Users should have the opportunity to assess the system before its implementation in their environment. To address this limitation, a prototype will be generated for user review during the implementation phase. This iterative feedback loop allows users to provide input, and adjustments can be made accordingly.

The decision to incorporate Prototyping is motivated by its capacity to rectify the inflexibility of Waterfall development, particularly in accommodating changing user requirements. The prototype will be developed as part of the implementation phase, presented to users for feedback before deployment. Modifications will be implemented based on users' insights.

Waterfall development has been chosen as the primary methodology due to its virtues of consistency and the thorough documentation it produces. The step-by-step approach of Waterfall development provides developers with clear guidelines for creating a fully functional system. Furthermore, the comprehensive documentation generated in the Waterfall model facilitates future system upgrades and serves as a valuable resource for system study and understanding.

4.2 Team Role and Responsibility

Supervisor(s):

Urooj Waheed
Yusra Mansoor

Team Members:

Name - **Ayushmann Kukreja**

Responsibilities - Testing, Requirement Gathering, Backend, Deployment

Name - **Murtaza Mehdi**

Responsibilities - Testing, Requirement Gathering, Backend, Deployment

Name - **Abdul Rehman Khan**

Responsibilities - Front-End

For the following major responsibilities, each member has equally contributed:-

R&D

Database

Documentation

4.3 Requirement Development

Stakeholder Analysis

To develop comprehensive requirements, it is essential to understand the needs and expectations of various stakeholders. Key stakeholders include:

- **Students:** Need a user-friendly platform to submit proposals, track progress, and receive feedback.
- **Supervisors:** Require efficient tools to manage multiple projects, monitor student progress, and provide timely feedback.
- **Industry Partners:** Seek a streamlined process to propose projects, collaborate with students, and monitor project outcomes.
- **Academic Administrators:** Need oversight capabilities to ensure compliance with institutional standards and streamline administrative tasks.

Functional Requirements

Centralized Communication

- **Messaging System:** Implement an integrated messaging system for real-time communication between students, supervisors, and industry partners.
- **Notification System:** Provide automated notifications for important updates, deadlines, and feedback.

Project Management

- **Proposal Submission:** Develop a module for students to submit project proposals, including templates and guidelines.
- **Progress Tracking:** Create a system for tracking project milestones, tasks, and deliverables.
- **Feedback Mechanism:** Enable supervisors to provide structured feedback on proposals, progress, and final submissions.

Standardized Workflows

- **Workflow Templates:** Introduce standardized workflows for project selection, progress tracking, and final evaluation.
- **Approval Process:** Implement a formal approval process for project proposals and significant milestones.

Student Support

- **Dashboard:** Develop a student dashboard to display project status, deadlines, and feedback.
- **Automated Reminders:** Set up automated reminders for upcoming deadlines and incomplete tasks.

Supervisor Support

- **Project Overview:** Provide a comprehensive overview of all supervised projects, including statuses and deadlines.
- **Mentorship Tools:** Implement tools to facilitate mentorship, such as meeting schedulers and document sharing.

Industry Collaboration

- **Partner Portal:** Create a dedicated portal for industry partners to propose projects, track progress, and collaborate with students.
- **Expertise Contribution:** Enable industry experts to contribute insights and resources to ongoing projects.

Non-Functional Requirements

Performance

- **Scalability:** Ensure the system can handle an increasing number of users and projects without performance degradation.
- **Reliability:** Design the system to be highly reliable, minimizing downtime and ensuring consistent availability.

Security

- **Data Protection:** Implement robust security measures to protect sensitive data, including encryption and access controls.
- **User Authentication:** Ensure secure user authentication and authorization mechanisms to prevent unauthorized access.

Usability

- **User-Friendly Interface:** Design an intuitive and easy-to-navigate interface for all user roles.
- **Accessibility:** Ensure the platform is accessible to users with disabilities, complying with relevant accessibility standards.

4.4 High Level Architecture

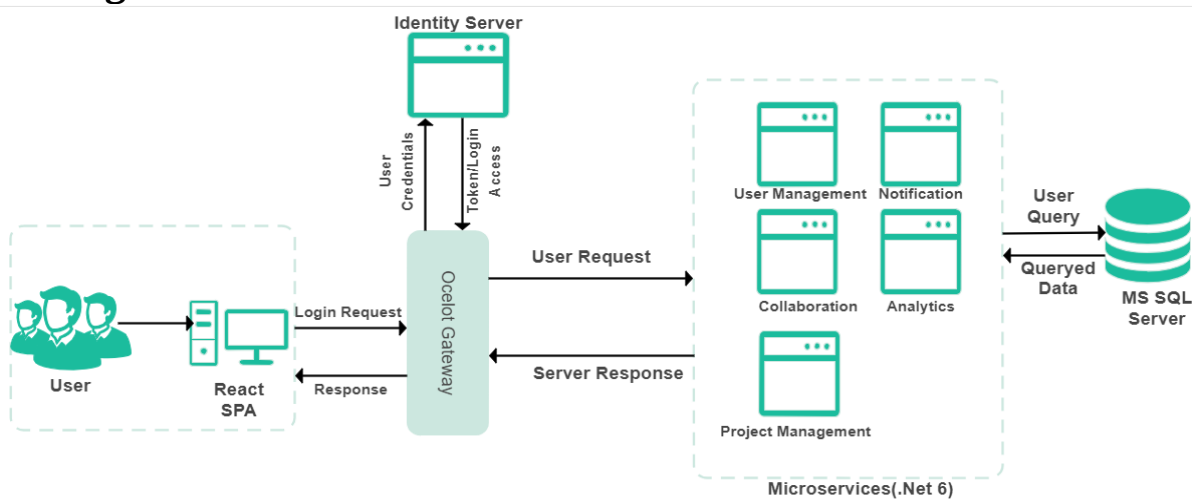


Figure 1: Context Diagram

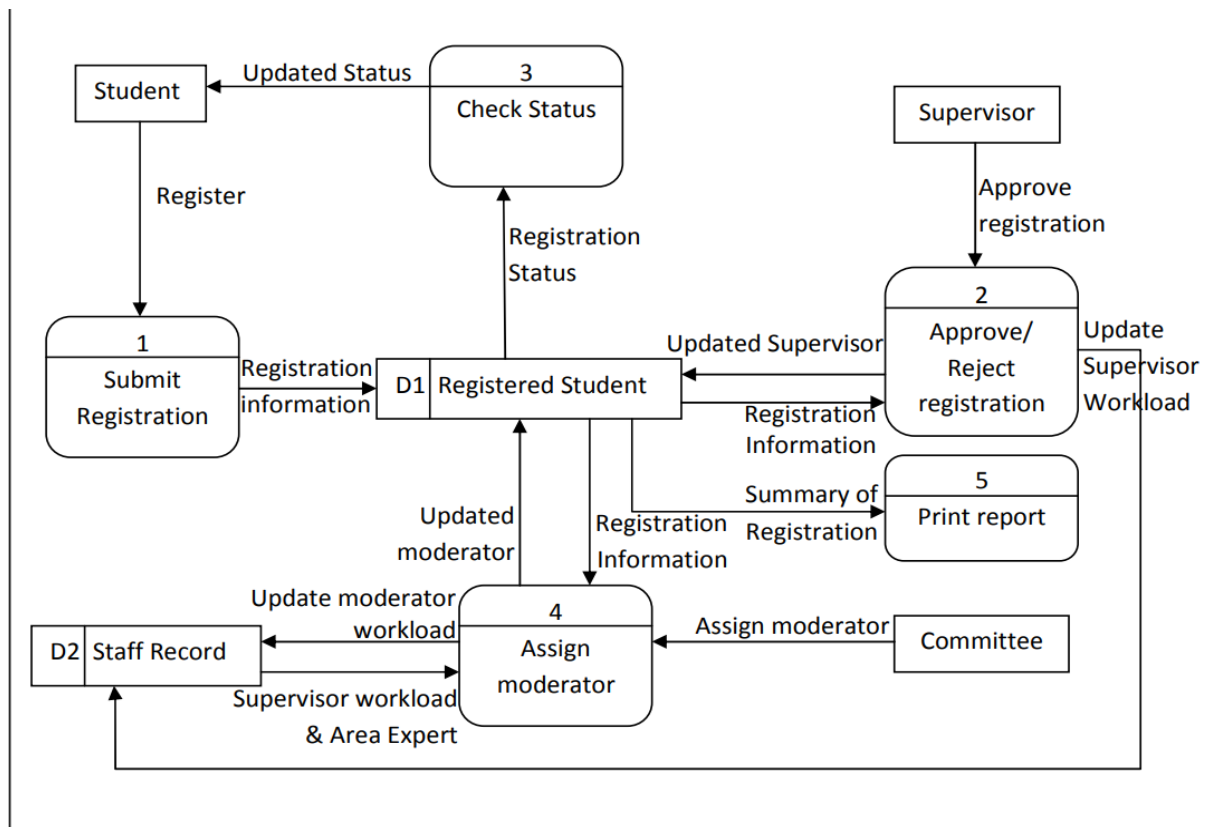


Figure 2: Level-0 Data Flow Diagram

Project Planning

5.1 Gantt Chart

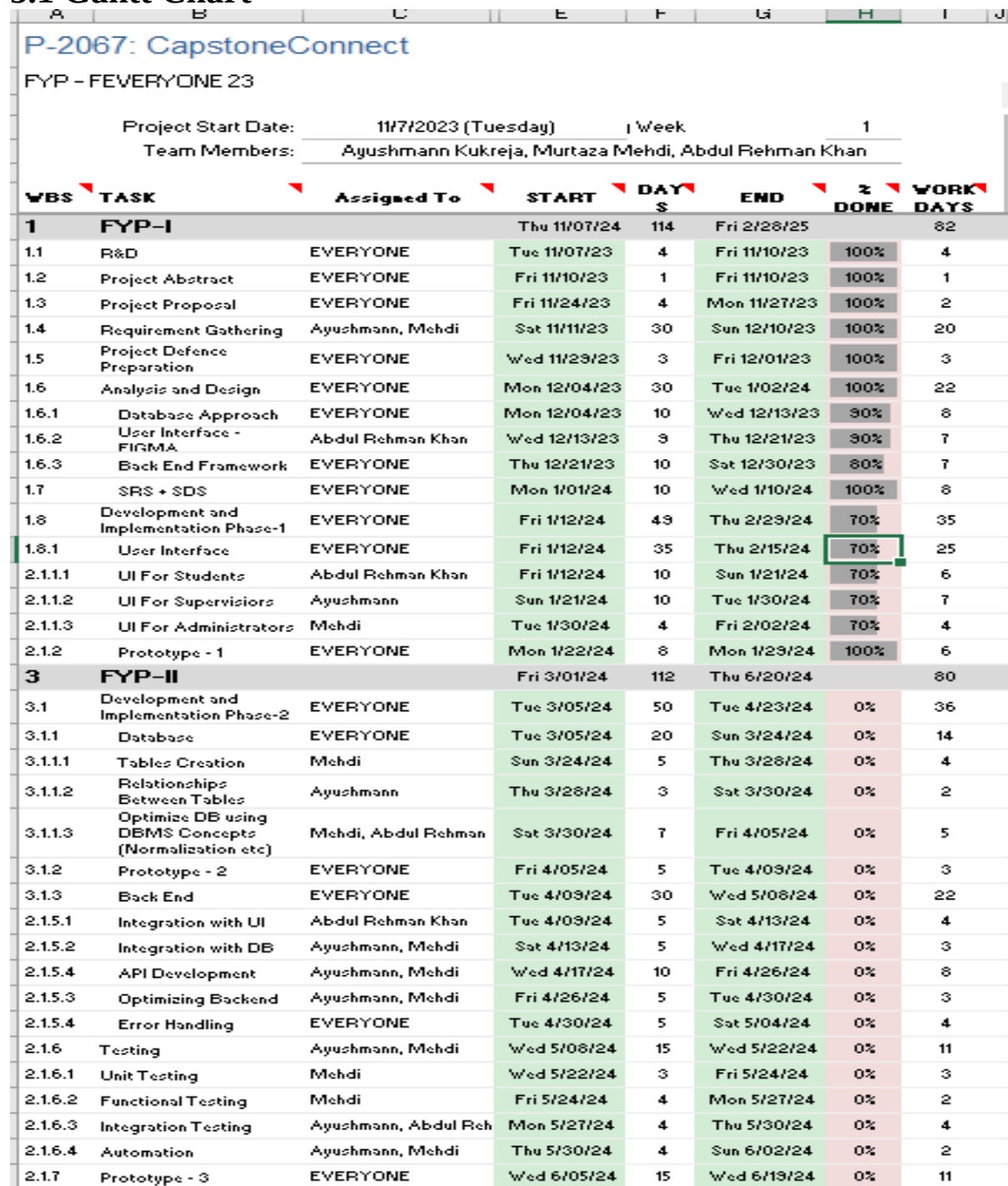


Figure 3: Gantt chart

6. Project Requirements

6.1 Software tools requirements:

Front End: React.js is a popular JavaScript library for building user interfaces, developed by Facebook.

Back End: ASP.NET Core is a versatile, open-source framework for building modern, cloud-based, and cross-platform applications.

Database: (SQL Server Management Studio)

IDE: Visual Studio 2022

Deployment: IIS

6.2 Hardware requirements

CPU: Opt for multicore processors like Intel Core i5/i7 or AMD Ryzen series for faster compilation and processing.

RAM: Ensure a minimum of 8GB RAM, with higher capacities (16GB or more) recommended for efficient multitasking and running development environments.

Storage: Priorities SSDs (Solid State Drives) to enhance read/write speeds, facilitating quicker software builds and overall system performance.

7 Budgeting/Costing

7.1 Estimated Budgeted Cost of the Project

1. Azure Services for Database and App Deployment:

- **Azure SQL Database:**
 - Standard Tier: \$15-\$30/month
- **Azure App Service:**
 - Standard Plan: \$75/month

Estimated Cost for 2 Months:

Service	Plan	Monthly Cost (USD)	Total Cost (2 Months) (USD)
SQL Database	Standard	\$30	\$60
App Service	Standard	\$75	\$150
Total			\$210

2. Professional Version of Visual Studio 2022:

- **Visual Studio 2022 Professional:**
 - Reduced Cost: 100,000 PKR

3. Development Cost:

- **Per Hour Rate:** 500 PKR
- **Total Hours:** 800

Total Development Cost:

800hours×500PKR/hour=400,000PKR

4. Summary of Costs:

Item	Cost (PKR)	Cost (USD)
Azure Services (2 months)	Approx. 63,000 PKR	(Approx. 210 USD)
Visual Studio 2022 Professional	100,000 PKR	-
Development (800 hours @ 500 PKR/hour)	400,000 PKR	-
Total Cost	563,000 PKR	-

Note:

- Exchange rate assumed: 1 USD = 300 PKR.
- Azure Services costs in USD are converted to PKR for total costing.

Total Cost (in PKR):

$$63,000\text{PKR (Azure Services)} + 100,000\text{PKR (Visual Studio)} + 400,000\text{PKR (Development)} = 563,000\text{PKR}$$

8. Project Deliverables:

8.1 Phase I - Alpha Prototype

The Alpha Prototype will be created by mid-January, which will contain interfaces of all the features (without implementation).

Expected Release: 01/23/2024

8.2 Phase II - Beta Prototype

The Beta Prototype will present detailed Database Working (Schemas, Tables, Relationships, and ERDs).

Expected Release: 2/18/2024

8.3 Phase III - Release Candidate

The Release Candidate will showcase a complete working application with all functionalities implemented on Backend.

Expected Release: 03/20/2024

8.4 Phase IV - Final Product

The Final Product will be delivered with complete documentation and deployment to all the affected members of the project.

Expected Release: 04/19/2024

9. Proposed GUI (Prototype)

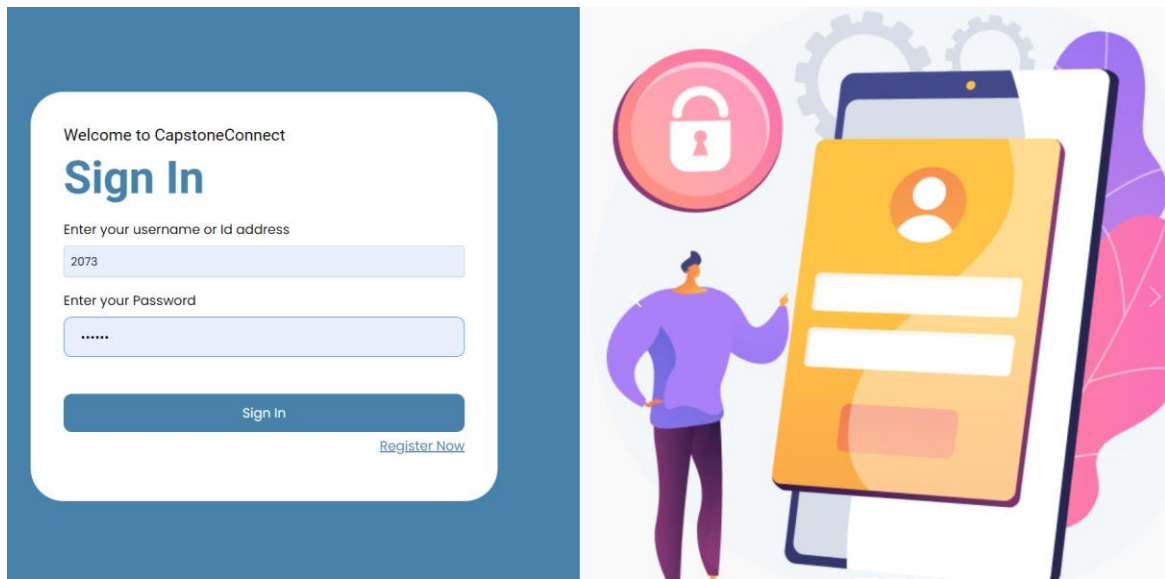


Figure 4: Login

The image shows a registration screen prototype for 'CapstoneConnect'. It features a white rounded rectangle on a blue background. At the top, it says 'Welcome to CapstoneConnect' and 'Sign Up'. Below this is a progress indicator with 'Step 1' and 'Step 2'. Under 'Step 1', there are three input fields: 'Select Number Of Group Members' with the value '2', 'Enter Project Title', and 'Enter Team Lead ID'. At the bottom of the form is a blue 'Next' button and a blue 'Back to login' link.

Figure 5: Registration

SUPERVISORS

Total Supervisors : 245 Available : 225 Occupied : 25

Supervisor	Subject	Availability
Supervisor 1	Network Engineering	3-01-2023 12:30 AM - 01:40 PM
Supervisor 2	BlockChain	3-01-2023 12:30 AM - 01:40 PM
Supervisor 3	Research Paper	3-01-2023 12:30 AM - 01:40 PM
Supervisor 4	AI & ML	3-01-2023 12:30 AM - 01:40 PM
Supervisor 5	Data Science	3-01-2023 12:30 AM - 01:40 PM
Supervisor 6	Robotics	3-01-2023 12:30 AM - 01:40 PM

Log Out

Figure 6: Available Supervisors

PROJECTS 09 - 01 - 2023 Search...

Ongoing	Completed	Missed
Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	Any task beyond Due Date Nothing to show
Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	
Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	Add a Quiz for Students of 3CO - JVV on the subject Fundamentals of Programming	

Log Out

Figure 7: Progress Tracking

PREVIOUS PROJECT REPOSITRIES

Overview Segments Dashboard

Search by project number, name, ... Filter

PROJECT NUMBER	PROJECT NAME	TEAM LEAD	SUPERVISOR	DESCRIPTION	
☐ 5146846548465	Jane Cooper	Jane Cooper	Jane Cooper	ILorem ipsum dolor sit amet consectetur adipisci...	View
☐ 5467319467348	Wade Warren	Wade Warren	Wade Warren	ILorem ipsum dolor sit amet consectetur adipisci...	View
☐ 1345705945446	Esther Howard	Esther Howard	Esther Howard	ILorem ipsum dolor sit amet consectetur adipisci...	View
☐ 544075497977	Cameron Williamson	Cameron Williamson	Cameron Williamson	ILorem ipsum dolor sit amet consectetur adipl...	View
☐ 1243467984543	Brooklyn Simmons	Brooklyn Simmons	Brooklyn Simmons	ILorem ipsum dolor sit amet consectetur adipl...	View
☐ 8454134649707	Leslie Alexander	Leslie Alexander	Leslie Alexander	ILorem ipsum dolor sit amet consectetur adipiscin...	View
☐ 2130164040451	Jenny Wilson	Jenny Wilson	Jenny Wilson	ILorem ipsum dolor sit amet consectetur adipisc...	View
☐ 0439104645404	Guy Hawkins	Guy Hawkins	Guy Hawkins	ILorem ipsum dolor sit amet consectetur adipiscin...	View

Log Out

Figure 8: Project Repository

FYP CALENDAR

FYP PROPOSAL EVALUATIONS

Proposal Submission Starts:	Sep 11, 2023
Proposal Submission Deadline:	Oct 21, 2023
Proposal Evaluation Starts:	Oct 9, 2023
Proposal Evaluation Deadline:	Oct 20, 2023

8TH SEMESTER

SRDS / Demo Submission Starts:	Sep 11, 2023
SRDS / Demo Submission Deadline:	Oct 21, 2023
Internal Evaluation 7th Starts:	Oct 9, 2023
Internal Evaluation 7th Deadline:	Oct 20, 2023

Log Out

Figure 9: Calendar

10 Meetings held with supervisor and/or client.

7/11/2023 - Idea was pitched by Supervisor to us. Ma'am Urooj Waheed explained how the current procedure of FYP is being done and how it could be improved.

10/11/2023 - This meeting was conducted by us, explaining our idea and vision about the project.

18/11/2023 - The purpose of this meeting was to gather requirements from university

References

- [1] R. I. H. N. B. a. H. M. Lounas, "An E-Collaboration Application For Final Year Project Management.," International Journal of e-Collaboration 17, 2023.
- [2] L. a. C. L. T. Chung-Hang, "The Development of Final Year Project Management System For Information Technology Programmes.," 2015.

A1B. COPY OF PROPOSAL EVALUATION COMMENTS BY JURY



PROJECT TITLE: CapstoneConnect
PROJECT STATUS: Strong Accepted
Proposal Defense: 5-March-2024

FYP-I Proposal Evaluation Feedback
Response Report

PROJECT CODE: P-2067
CS - FALL – 2024



	Jury Revision Comments	Response	Action	Supervisor Signature
1	Need to work on scope. Scope is small as per understanding. Project presentation and document Needs to be improved.	Agreed	Expanded Project Scope: Implemented similarity detection mechanism for project evaluation. Improved Presentation and Documentation: Enhanced quality and thoroughness of project materials.	

A2. REQUIREMENT SPECIFICATIONS

- 1. Introduction**
 - 1.1 Purpose of Document**
 - 1.2 Intended Audience**
- 2. Overall System Description**
 - 2.1 Project Background**
 - 2.2 Problem Statement**
 - 2.3 Project Scope**
 - 2.4 Not In Scope:**
 - 2.5 Project Objectives**
 - 2.6 Stakeholders & Affected Groups**
 - 2.7 Operating Environment**
 - 2.8 System Constraints**
 - 2.9 Assumptions & Dependencies**
- 3. External Interface Requirements**
 - 3.1 Hardware Interfaces**
 - 3.2 Software Interfaces**
 - 3.3 Communications Interfaces**
- 4. System Functions / Functional Requirements**
 - 4.1 System Functions**
 - 4.2 Use Cases**
 - 4.2.1 List of Actors**
 - 4.2.2 List of Use Cases**
 - 4.2.3 Use Case Diagram**
 - 4.2.4 Description of Use Cases**
- 5. Non - Functional Requirements**
 - 5.1 Performance Requirements**
 - 5.2 Safety Requirements**
 - 5.3 Security Requirements**
 - 5.4 Reliability Requirements**
 - 5.5 Usability Requirements**
 - 5.6 Supportability Requirements**
 - 5.7 User Documentation**
- 6. References**

1 Introduction

1.1 Purpose of Document

The purpose of the SRS document is to provide a detailed and comprehensive description of the functional and non-functional requirements of the CapstoneConnect portal. The SRS document will serve as a contract between the developers and the stakeholders, ensuring a clear and mutual understanding of the scope, features, and specifications of the project. The SRS document will also guide the design, development, testing, and deployment of the CapstoneConnect portal, facilitating a smooth and efficient project execution.

1.2 Intended Audience

The intended audiences of the SRS document are:

- The developers who will design, implement, test, and deploy the CapstoneConnect portal based on the requirements specified in the SRS document.
- The stakeholders who will review, approve, and provide feedback on the SRS document and the CapstoneConnect portal.
- The users who will use the CapstoneConnect portal for their capstone projects.

2 Overall System Description

2.1. Project Background

Final Year Projects (FYPs) serve as a critical component of academic programs, allowing students to synthesize and apply their knowledge in a practical, real-world context. These projects often involve complex problem-solving, research, and collaboration, requiring effective management to ensure successful outcomes. However, traditional FYP management systems are often fragmented, relying on disjointed communication channels, manual tracking methods, and inconsistent processes. This can lead to challenges for students in submitting proposals, tracking progress, and receiving timely feedback. Similarly, supervisors struggle to efficiently manage multiple projects and provide adequate guidance. Recognizing these issues, there is a pressing need for a streamlined, centralized solution to enhance the FYP experience.

2.2 In Scope

The scope of the CapstoneConnect project is carefully defined to encompass the specific functionalities, features, and objectives intended for development and implementation. The project's in scope elements are outlined below:

- **User Registration and Authentication:**

Develop a secure and user-friendly registration process for students, supervisors, and administrators. Implement robust authentication mechanisms to ensure secure access to CapstoneConnect.

- **Project Proposal Submission:**

Create a streamlined and intuitive interface for students to submit comprehensive project proposals. Develop a proposal management system that allows supervisors to review and provide feedback on project proposals.

- **Progress Tracking and Milestone Management:**

Design a centralized dashboard for students and supervisors to track project progress.

Implement milestone management features to enable the setting and monitoring of project milestones throughout the FYP lifecycle.

- **Communication Hub:**

Establish a communication hub within CapstoneConnect for seamless interaction between students and supervisors.

Include messaging and notification features to facilitate timely communication and updates.

- **Documentation and Resource Management:**

Create a structured system for the submission and storage of project-related documents.

Develop resource management tools to facilitate the sharing of relevant materials among students and supervisors.

Collaborative Tools:

Integrate collaborative tools, such as discussion forums or project-specific chat features, to encourage knowledge sharing among students and supervisors.

- **Evaluation and Grading System:**

Implement a comprehensive evaluation and grading system for supervisors to assess and provide feedback on project submissions.

Ensure transparency in the grading process and enable students to access their evaluation results.

Analytics and Reporting:

Develop analytics and reporting functionalities to generate insights into project trends, user engagement, and overall system performance.

Provide administrators with tools to monitor the impact of CapstoneConnect on academic outcomes.

- **Security Measures:**

Incorporate robust security measures to protect user data and ensure compliance with data protection and privacy regulations.

Regularly conduct security audits and vulnerability assessments to maintain a secure environment.

- **Scalability and Integration:**

Design CapstoneConnect to be scalable, accommodating potential growth in the number of users and projects.

Collaborate with the IT department to ensure seamless integration with existing institutional systems.

- **User Training and Support:**

Develop comprehensive training materials for students, supervisors, and administrators.

Provide ongoing user support, including helpdesk services and documentation.

- **Continuous Improvement and Feedback Loop:**

Establish a structured feedback loop to gather input from users for iterative improvements.

Plan for regular updates and enhancements based on user feedback and evolving needs.

2.3 Out of Scope

While the CapstoneConnect project focuses on specific functionalities and objectives to enhance Final Year Project (FYP) management, there are certain elements that fall outside the defined scope. The out scope elements are outlined below:

Course Management:

CapstoneConnect does not extend to the management of regular coursework or other academic programs outside the scope of FYPs.

Grades and Transcripts:

CapstoneConnect is not responsible for managing or recording grades for courses outside the FYP or generating academic transcripts.

Student Enrollment:

The platform does not handle the enrollment process for courses or academic programs; it specifically focuses on FYP management.

Student Recruitment:

The project does not include features for student recruitment, admission processes, or enrollment in academic programs.

General Communication Platform:

While CapstoneConnect includes project-specific communication tools, it does not serve as a general communication platform for broader institutional or non-project-related communications.

2.4 Project Objectives

The objectives of CapstoneConnect are to streamline and enhance the entire Final Year Project process, benefiting all stakeholders involved:

1. **Centralize Communication:** Develop a platform that centralizes communication for students, supervisors, and industry partners, reducing misunderstandings and improving collaboration.
2. **Automate Tracking and Management:** Implement automated systems for tracking project proposals, progress, and feedback to increase accuracy and efficiency, reducing the time and effort required for manual tracking.
3. **Standardize Processes:** Establish standardized workflows for project selection, progress tracking, and feedback to ensure consistency and clarity throughout the FYP process.
4. **Enhance Student Experience:** Create a user-friendly interface with automated reminders and easy access to project details, helping students stay organized and on track with their projects.
5. **Support Supervisors:** Provide tools for efficient project management, enabling supervisors to oversee multiple projects effectively and focus more on providing quality mentorship.
6. **Facilitate Industry Collaboration:** Enable better interaction with industry partners through a platform that allows for easy monitoring of project progress and contribution of expertise, enhancing the practical relevance of capstone projects.

2.5 Stake Holders

1. **Students (Primary Stakeholders):-**

The primary beneficiaries of the CapstoneConnect project are the students undertaking Final Year Projects (FYPs). They will use the platform for proposal submission, progress tracking, and communication with supervisors.

2. **Supervisors (Primary stakeholders):-**

Faculty members serving as project supervisors play a critical role in guiding and evaluating students throughout the FYP process. They will use CapstoneConnect to monitor progress, provide feedback, and assess project submissions.

3. **Administrators (Key stakeholders):-**

Institutional administrators oversee the overall academic processes and ensure the successful integration of CapstoneConnect into existing systems.

2.6 Operating Environment

The CapstoneConnect portal will operate as a web-based application that can be accessed through any standard web browser, such as Microsoft Edge, Google Chrome, Mozilla Firefox, or Safari. The portal will be hosted on a cloud server, such as Microsoft Azure or Amazon Web Services, and will use a relational database management system, such as MySQL or PostgreSQL, to store and retrieve data.

2.7 System Constraints

System constraints for CapstoneConnect:

1. **Compatibility:** Ensure compatibility across browsers, OS, and devices for a consistent user interface.
2. **Security:** Prioritize data and communication security, preventing unauthorized access or modification.
3. **Performance:** Maintain fast and reliable functionality, handling high traffic without compromising user experience.
4. **Scalability:** Accommodate user and data growth, adjusting system resources accordingly.
5. **Maintainability:** Ensure ease of updates, modifications, and debugging while adhering to coding standards.

2.8 Assumptions and Dependencies

Assumptions:

1. **User Engagement:** Users will actively engage with training and documentation to familiarize themselves with CapstoneConnect for effective implementation.
2. **Institutional Support:** Strong institutional support, including administrator backing and commitment to system integration, is assumed for successful deployment.
3. **Data Accuracy:** Users will provide accurate information during proposal submission and project tracking for reliable progress tracking within CapstoneConnect.
4. **User Feedback Incorporation:** Users will actively provide constructive feedback on the CapstoneConnect platform for continuous improvement.

3 External Interface Requirements

3.1 Hardware Interfaces

As the application is designed to operate over the internet, all necessary hardware for establishing internet connectivity will serve as the system's hardware interfaces. Examples of such interfaces include modems, WAN–LAN connections, and Ethernet cross-cables.

3.2 Software Interfaces

1) **Database Management System (DBMS):**

Name of Application: SQL Server

External Owner: Database Administrator

Interface Details: Interaction involves SQL queries and data retrieval methods. The Database schema and connection details will need to be coordinated with the Database Administrator.

2) **Authentication Service:**

Name of Application: OAuth 2.0

External Owner: Identity Provider

Interface Details: Integration involves OAuth 2.0 protocols for secure user authentication. Coordination with the Identity Provider is needed for API keys, client IDs, and secret configurations.

3) **Email Service:**

Name of Application: SendGrid

External Owner: Email Service Provider

Interface Details: The FYP portal interfaces with SendGrid for sending email notifications. Coordination involves API key setup, email templates, and delivery configurations with the Email Service Provider.

3.3 Communications Interfaces

Describe communications interfaces to other systems or devices, such as local area networks for the intranet communication will be through TCP/IP protocol suite.

4 Functional Requirements

4.1 Functional Hierarchy

5 Function Category	Meaning
<i>Evident</i>	<i>Should perform, and user should be cognizant that it is performed.</i>
<i>Hidden</i>	<i>Should perform, but not be visible to users. This is true of many underlying technical services, such as save information in a persistent storage mechanism. Hidden functions are often missed during the requirements gathering process.</i>

Ref #	Functions	Category	Attribute	Details & Boundary Constraints
R1	Display a list of pervious FYPs	Evident	Information about previous FYPs.	Users will be able to see a repository of previous projects. Details of what is to be shown depends on the admin of the portal.
R2	Facilitate communication between users and supervisors	Evident	Communication platform.	A way for students and supervisors to communicate. Students can approach supervisors with a request to be supervised by them. If the supervisor accepts the request they can chat with each other from that point on.
R3	Improve documentation processes by providing an organized platform	Evident	Documentation processes	Improving the documentation process by providing a framework to document your tasks. The portal will keep a record of every entry
R4	Integrate progress tracking tools to enable real-time monitoring of project milestones	Evident	Progress tracking tools	Improving the progress tracking by the use of a progress tracking tool. Constraint: An external project tracking tool will be integrated that will require subscription.
R5	Implement a comprehensive evaluation and grading system	Evident	Grading system	Supervisors will be able to grade students on the base of their work.
R6	Display a list of available Supervisors	Evident	Available Supervisors	Students will be able to view the lists of supervisors available for FYP.

System Attributes/ Nonfunctional Requirements

Attribute	Details and Boundary Constraints	Category
<i>Intuitive and user-friendly interface</i>	<i>None specified.</i>	<i>Mandatory</i>
<i>High system reliability to ensure minimal downtime.</i>	<i>None specified.</i>	<i>Mandatory</i>
<i>Robust security measures to protect user data and prevent unauthorized access.</i>	<i>Compliance with data protection and privacy regulations.</i>	<i>Mandatory</i>
<i>High system availability to ensure users can access the system when needed.</i>	<i>None specified</i>	<i>Mandatory</i>
<i>Integration of a user feedback mechanism for continuous improvement.</i>	<i>None specified</i>	<i>Optional</i>
<i>Compatibility with different web browsers, operating systems, and devices.</i>	<i>None specified</i>	<i>Mandatory</i>

4.2 Use cases

4.2.1 List of Actors

1. Student:

Enrolled FYP course user. Can browse, select, and submit proposals, communicate with supervisors, upload documents, track progress, and receive feedback.

2. Supervisor:

Faculty or industry user overseeing projects. Can view, approve/reject proposals, communicate with students, monitor progress, and evaluate project submissions.

3. Administrator:

Staff or coordinator managing FYP course and portal. Can create/edit/delete projects, assign/match supervisors, manage user profiles, generate reports, and configure portal settings.

4.2.2 List of Use Cases

1. View Supervisors and sends request:

Students view supervisors available for FYP and sends request for supervision for FYP.

2. Previous Projects Repository:

Students and supervisor view previous FYPs with titles, descriptions, supervisors, and status.

3. Track FYP Progress:

Students and Supervisors track FYP progress, upload documents (literature review, design, code, etc.).

4. Evaluate and Grade Project:

Supervisor evaluates and grades FYP submissions using predefined criteria.

5. Communication:

Users (students/supervisors) communicate through messages related to FYP

4.2.3 Use Case Diagram

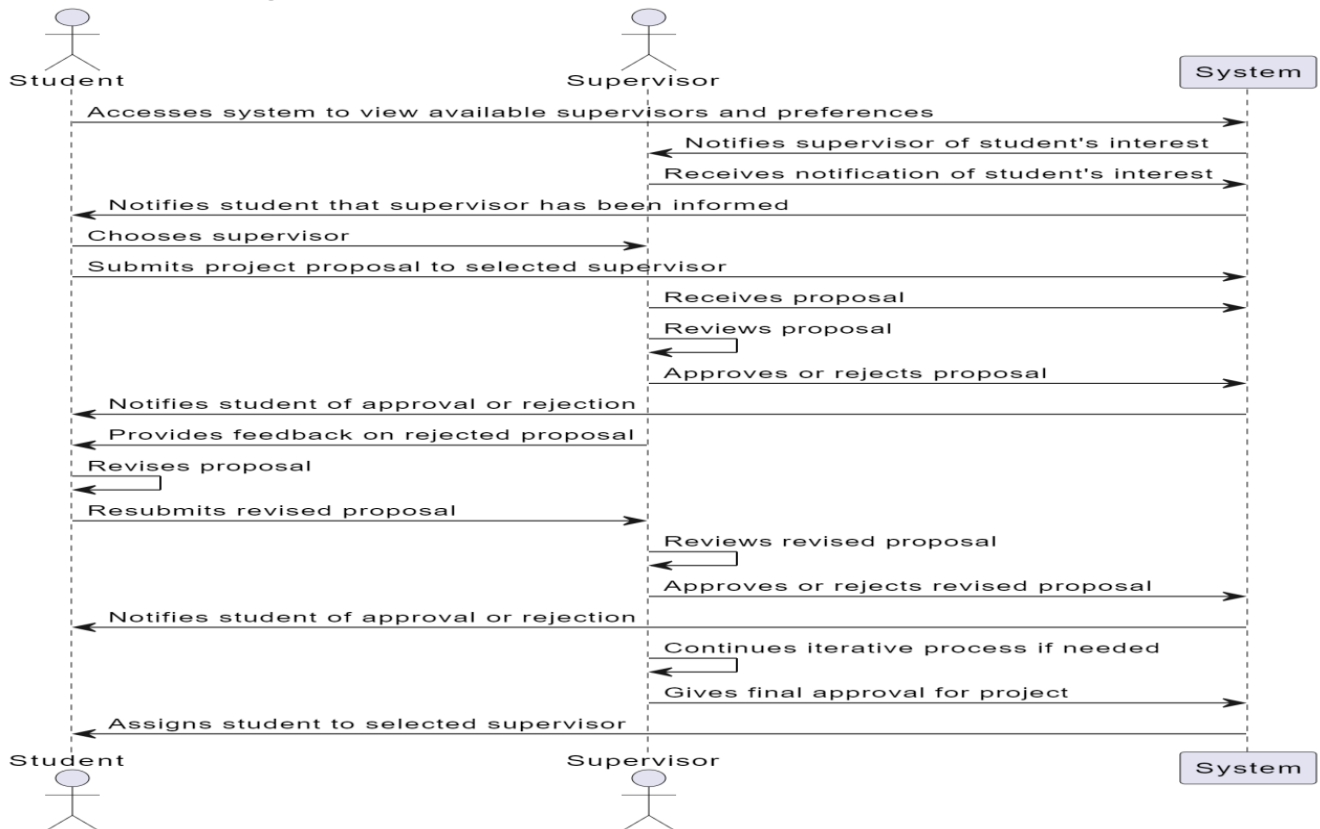


Figure 10: User Case 1: View Supervisors

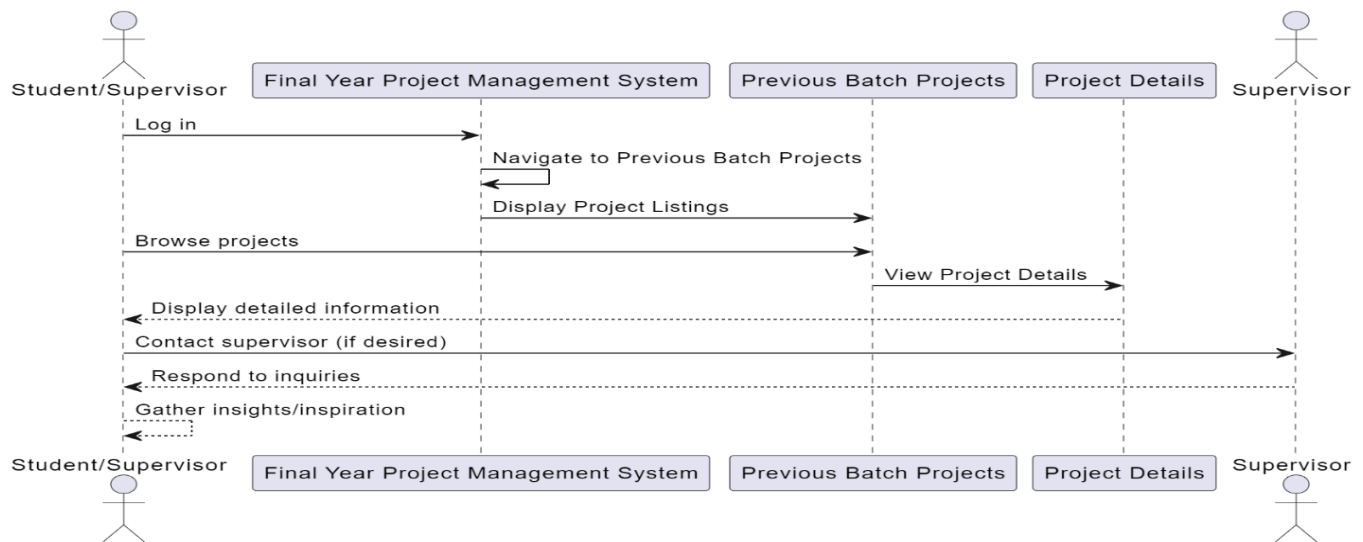


Figure 11: Use Case 2: Previous Projects Repository

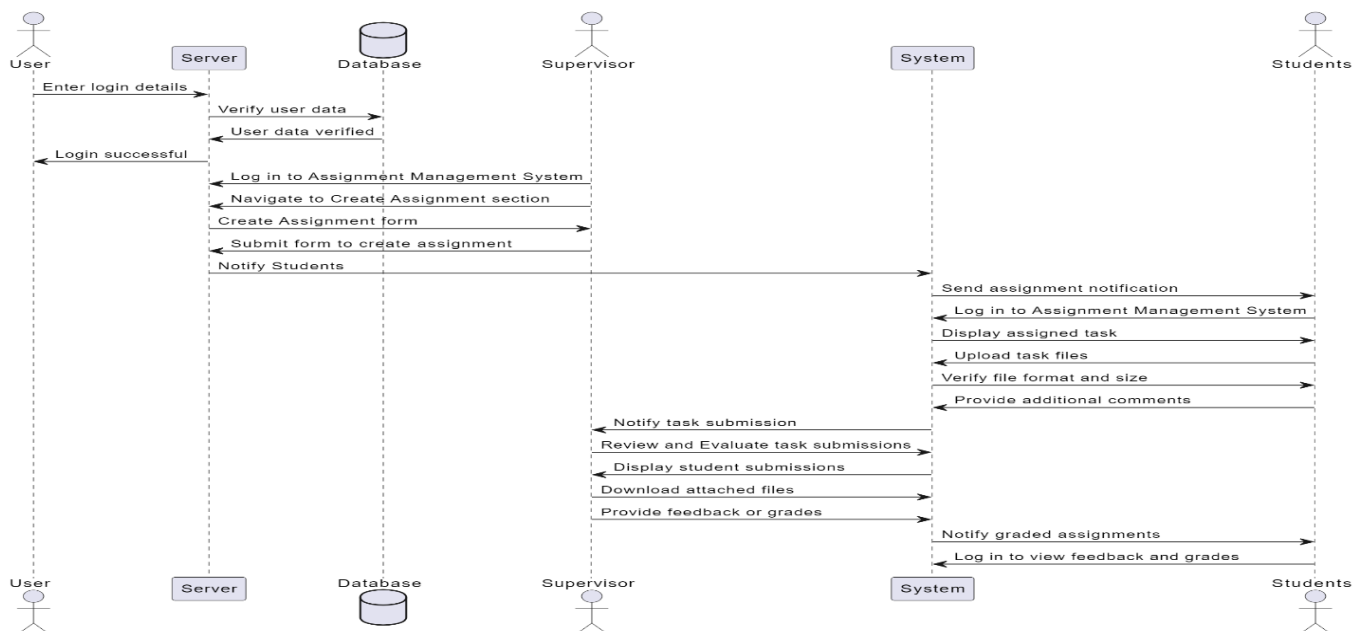


Figure 12: Use Case 3&4: Track and Document Progress, Evaluate and Grade Project

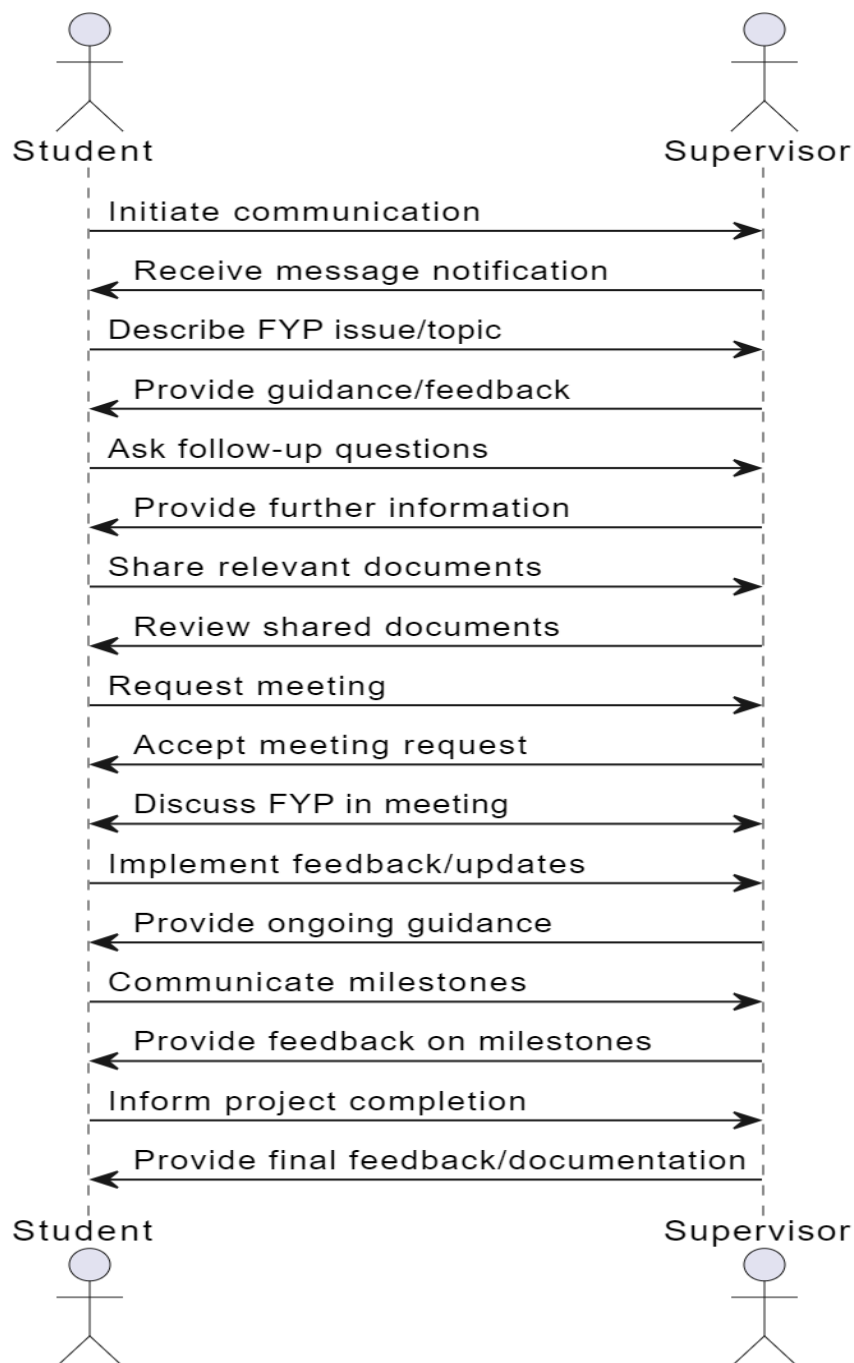


Figure 13: Use Case 5: Communication

4.2.4 Description of Use Cases

Section: Main		
Name:		View Supervisors and sends request
Actors:		Students and Supervisors
Purpose:		Student views available supervisors for FYP on the portal and approaches supervisors
Description:		A student accesses the portal to view available supervisors and their preferences.
Cross References:		Functions: R6 Use Cases: Available Supervisors will be displayed
Pre-Conditions		User must be logged in.
Successful Post-Conditions		Users can view available supervisors on the portal and choose their supervisor.
Failure Post-Conditions		Users cannot access or view available supervisor.
Typical Course of Events		
Actor Action		System Response
1	The use case begins when a user logs into the FYP portal.	Users are validated and allowed to use the portal.
2	Students initiate a call to view available supervisor and send request to supervisor with their idea	The system displays a list of available supervisor
3	Supervisors provide response (approval or rejection)	The system notifies student about supervisor's decision

Section: Main		
Name:		<i>Previous Projects Repository</i>
Actors:		<i>Students, Supervisors</i>
Purpose:		<i>View previous projects on the portal.</i>
Description:		<i>A student accesses the portal to browse available FYPs. The system displays FYP details, including titles, descriptions, assigned supervisors, and project status. Students can take an idea of how these projects were developed</i>
Cross References:		<i>Functions: R1</i> <i>Use Cases: Project repository will be displayed</i>
Pre-Conditions		<i>User must be logged in.</i>
Successful Post-Conditions		<i>Users can view information about previous FYPs according to the admin's specifications.</i>
Failure Post-Conditions		<i>Users cannot access or view information about previous FYPs.</i>
Typical Course of Events		
Actor Action		System Response
1	<i>The use case begins when a user logs into the FYP portal.</i>	<i>Users are validated and allowed to use the portal.</i>
2	<i>Users initiate a call to the projects repository section.</i>	<i>The system displays a list of previous FYPs, presenting details based on admin configurations</i>

Section: Main		
Name:		Track FYP Progress
Actors:		Students, Supervisors
Purpose:		Keep track of FYP
Description:		A student regularly updates their task or assignment which makes it easy for both student and supervisor to keep track of FYP progress.
Cross References:		Functions: R3&R4 Use Cases: Track FYP Progress
Pre-Conditions		Student must have gotten approval for their project proposal
Successful Post-Conditions		Students and supervisors can keep track of FYP progress
Failure Post-Conditions		Students and supervisors cannot keep track of FYP progress
Typical Course of Events		
Actor Action		System Response
1	The use case begins when a user logs into the FYP portal.	Users are validated and allowed to use the portal.
2	Supervisors upload a task for student	The system notifies students that the supervisor has upload a task.
3	Student works on the task and updates the status of task	The system notifies supervisor which task is done and request for feedback and grading

Section: Main		
Name:		<i>Evaluate and Grade Project</i>
Actors:		<i>Supervisors</i>
Purpose:		<i>Supervisors regularly grade and evaluate tasks.</i>
Description:		<i>A supervisors provide regular updates and grade students submission.</i>
Cross References:		<i>Functions: R5</i> <i>Use Cases: Evaluate and Grade Project</i>
Pre-Conditions		<i>Supervisor must be logged In and Student must have updated their submissions.</i>
Successful Post-Conditions		<i>Student gets notification about feedback from supervisor.</i>
Failure Post-Conditions		<i>Student don't get any feedback from supervisors.</i>
Typical Course of Events		
Actor Action		System Response
1	<i>The use case begins when a supervisor logs into the FYP portal.</i>	<i>Users are validated and allowed to use the portal.</i>
2	<i>Supervisor initiate a call to the view a submissions of student.</i>	<i>The system displays submission of student</i>

Section: Main		
Name:		Communication
Actors:		Supervisors and Students
Purpose:		Students and Supervisors communication
Description:		The CapstoneConnect portal bridges the gap between student and supervisor by providing a platform for communication.
Cross References:		Functions: R2 Use Cases: Communication
Pre-Conditions		Supervisor and Students are available for communication
Successful Post-Conditions		Student and Supervisor communicate and deliver their message through this platform.
Failure Post-Conditions		Both actors aren't able to able to deliver their message.
Typical Course of Events		
Actor Action		System Response
1	The use case begins when user logs into the FYP portal.	Users are validated and allowed to use the portal.
2	Student sends a message to supervisor related to FYP.	The system sends a notification about message to supervisor
3	Supervisors replies back to student regarding their query	The system sends a response back to student.

5 Non Functional Requirements

5.1 Performance Requirements

The CapstoneConnect portal should provide fast and reliable functionality, and should handle high traffic and concurrent requests without compromising the user experience or the system stability. The portal should also have a low response time and a high throughput for all the system functions.

5.2 Safety Requirements

The CapstoneConnect portal should ensure the safety and integrity of the user data and the communication between the users and the server, and should prevent data loss, corruption, or leakage. The portal should also have a backup and recovery mechanism to restore the system in case of a failure or a disaster.

5.3 Security Requirements

The CapstoneConnect portal should ensure the security and privacy of the user data and the communication between the users and the server, and should prevent unauthorized access, modification, or deletion of the data. The portal should implement security measures, such as encryption, authentication, authorization, and auditing.

5.4 Reliability Requirements

The CapstoneConnect portal should ensure the reliability and availability of the system functions and the user data, and should minimize the occurrence and impact of errors, faults, or failures. The portal should also have a high mean time between failures (MTBF) and a low mean time to repair (MTTR).

5.5 Usability Requirements

The CapstoneConnect portal should ensure the usability and accessibility of the system functions and the user interface, and should meet the user expectations and satisfaction. The portal should also follow the usability principles and best practices, such as simplicity, consistency, feedback, and error prevention. The portal should also support different languages, cultures, and preferences of the users.

5.6 Supportability Requirements

The CapstoneConnect portal should ensure the supportability and maintainability of the system functions and the user data, and should facilitate the update, modification, and debugging of the system. The portal should also follow the coding standards and best practices, such as modularity, readability, and documentation. The portal should also provide technical support and customer service to the users.

5.7 User Documentation

The CapstoneConnect portal should provide user documentation that describes the system functions, features, and specifications, and guides the users on how to use the system effectively and efficiently. The user documentation should also include tutorials, FAQs, and troubleshooting tips. The user documentation should be clear, concise, and accurate, and should be updated regularly.

6 References

- [1] R. I. H. N. B. a. H. M. Lounas, "An E-Collaboration Application For Final Year Project Management.," International Journal of e-Collaboration 17, 2023.
- [2] L. a. C. L. T. Chung-Hang, "The Development of Final Year Project Management System For Information Technology Programmes.," 2015.

A3. DESIGN SPECIFICATIONS

1. **Introduction**

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1.3 *Document Convention*

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5. **References**

1 Introduction

1.1 Purpose of Document

This document outlines the design of the Capstone Connect project, a Final Year Project (FYP) portal simulator. It provides detailed information on the expected input, output, classes, and functions, as well as the interactions between these elements to meet the desired requirements. At the conclusion of the document, you will find detailed figures illustrating the relationships and flow of information between different classes.

1.2 Intended Audience

The intended audiences of the SDS document are:

- **The developers** who will design, implement, test, and deploy the CapstoneConnect portal based on the requirements specified in the SDS document.
- **The stakeholders** who will review, approve, and provide feedback on the SDS document and the CapstoneConnect portal.
- **The users** who will review, approve, and provide feedback on the SDS document and the CapstoneConnect portal.

1.3 Document Convention

Font Family – Times New Roman

Font Sizes:

Top Heading – 20

Middle Headings – 16

Internal Headings – 14

Text - 12

1.4 Project Overview

The Capstone Connect portal aims to address the existing challenges in the academic capstone project lifecycle by creating a centralized and efficient platform. This platform will streamline project selection, enhance collaboration, and improve communication among students, faculty, and industry stakeholders.

1.5 Scope

Project scope definition is the process whereby a project is defined and prepared for execution. It helps to decide on whether or not to proceed with the project. The scope of the CapstoneConnect project is carefully defined to encompass the specific functionalities, features, and objectives intended for development and implementation. The projects in scope elements are outlined below:

1. User Authentication:

Develop secure and user-friendly registration for students, supervisors, and administrators with robust authentication mechanisms.

2. Progress Tracking and Milestone Management:

Design a centralized dashboard for tracking project progress; implement milestone management for FYP lifecycle monitoring.

3. Communication Hub:

Establish a seamless communication hub with messaging and notifications between students and supervisors.

4. Documentation and Resource Management:

Create a structured system for project document submission and storage; develop tools for resource sharing.

5. Collaborative Tools:

Integrate discussion forums and chat features to encourage knowledge sharing among users.

6. Evaluation and Grading System:

Implement a comprehensive system for supervisors to assess and provide transparent feedback on project submissions.

7. Analytics and Reporting:

Develop analytics for project trends, user engagement, and system performance; provide admin tools for monitoring.

8. Scalability and Integration:

Design for scalability, accommodating user and project growth; ensure seamless integration with institutional systems.

9. User Training and Support:

Develop training materials for users and provide ongoing support, including helpdesk services.

10. Continuous Improvement and Feedback Loop:

Establish a structured feedback loop for iterative improvements based on user input and evolving needs.

2 Design Considerations

2.1 *Assumptions and Dependencies*

Assumptions:

1. CapstoneConnect portal will use a web-based platform compatible with the chosen software architecture.
2. Compatibility with major browsers like Microsoft Edge, Google Chrome, and Mozilla Firefox is ensured.
3. The portal will feature a consistent and intuitive user interface adhering to UX design best practices.
4. A secure and efficient database system will store and retrieve data accurately.
5. The authentication system will be robust and flexible, accommodating various user types and roles.
6. Comprehensive documentation will support development, testing, and maintenance.

Dependencies:

1. Reliance on web server and network availability and performance.
2. Quality and completeness of requirements provided by stakeholders are crucial.
3. User and supervisor cooperation and feedback are essential throughout the project.
4. Integration and compatibility with external systems (institutional systems, email service, discussion forum) are dependencies.
5. Adherence to data protection and security regulations is vital.

2.2 Risks and Volatile Areas

1. User Feedback

User Feedback is crucial for system improvement, yet it introduces the risk of conflicting requirements. To address this, the system will employ a feedback loop for continuous user monitoring, emphasizing prioritization and validation before integration into the design.

2. Technology Updates

Technology Updates are inevitable and may impact design aspects such as compatibility and security. Mitigation involves using well-established, supported technologies, adhering to coding best practices, and having a contingency plan for potential technology migrations.

3. Security Threats

Security Threats pose a constant risk, necessitating robust measures like encryption, access controls, and regular audits. The system will also have a contingency plan for swift recovery from security breaches.

4. Scalability Challenges

Scalability Challenges may arise with system growth. Mitigation involves designing a scalable architecture, utilizing cloud-based services for flexibility, and closely monitoring and optimizing system performance and resource utilization.

3 System Architecture

3.1 System Level Architecture

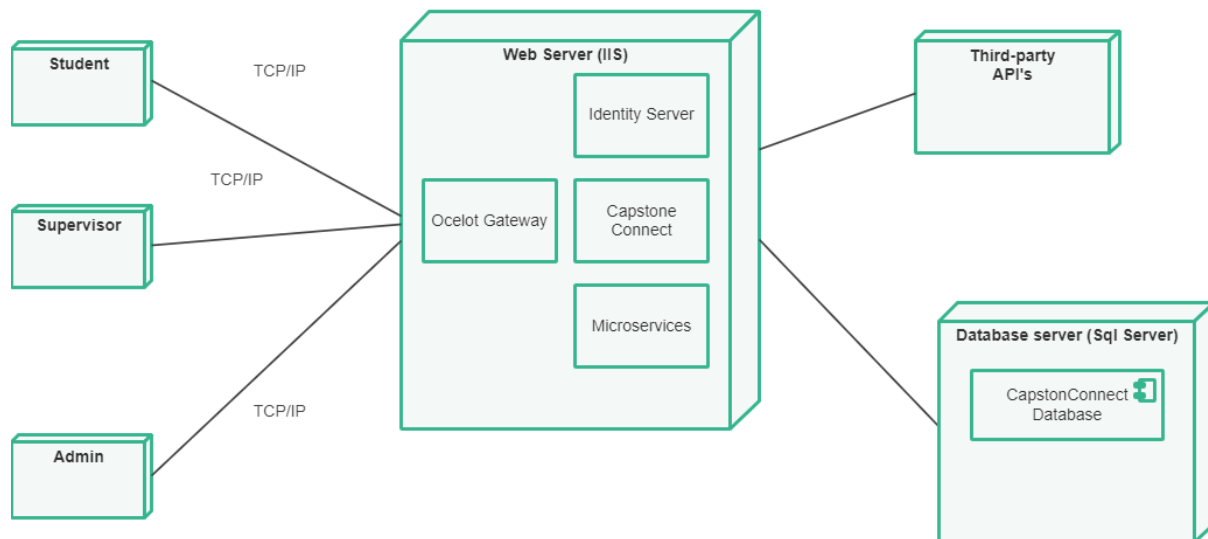


Figure 14: System Level Architecture

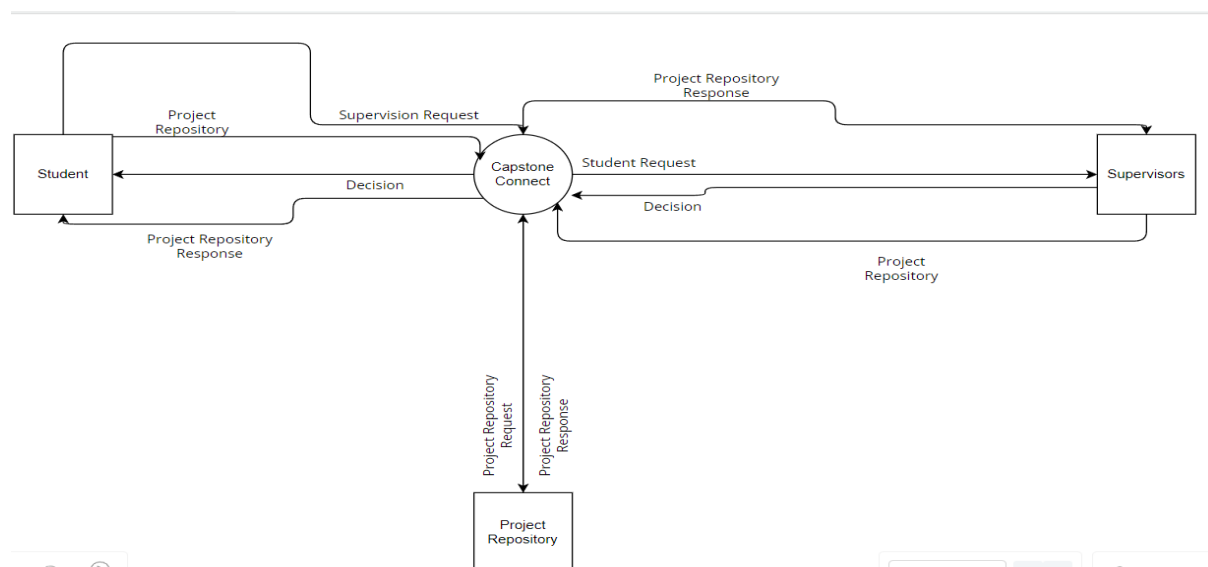


Figure 15: System Level Architecture

3.2 Software Architecture

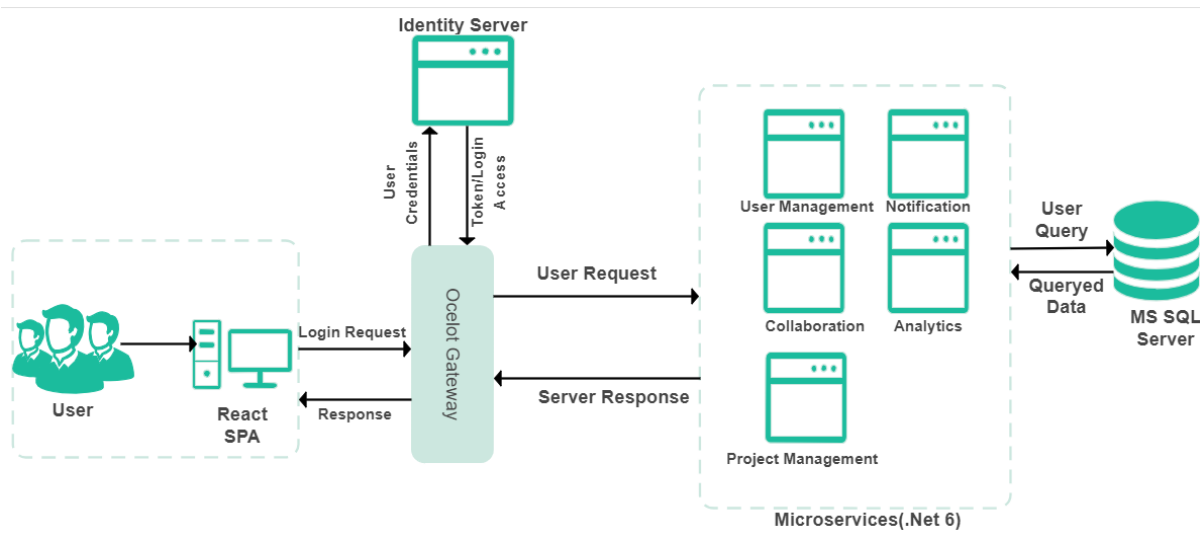


Figure 16: Software Architecture

4 Design Strategy

1 User-Centered Design:

The portal should be designed with the user in mind, taking into consideration their needs and concerns. This includes making the portal easy to navigate and understand and ensuring that the information is presented clearly and concisely.

2 Security and Privacy:

The portal should be designed with security and privacy in mind, to protect users' personal information and prevent unauthorized access to the portal.

3 Mobile-Friendly:

The portal should be designed to be mobile-friendly, as many users will likely access it using their smartphones or other mobile devices.

4 Feedback and Improvements:

The portal should be designed with the ability to collect feedback and make improvements, based on user feedback and usage data.

5 Ease of Use:

The portal should be designed to be simple and easy to use, with clear Instructions and providing useful information and resources throughout the process.

5 Detailed System Design

5.1 Design Class Diagram

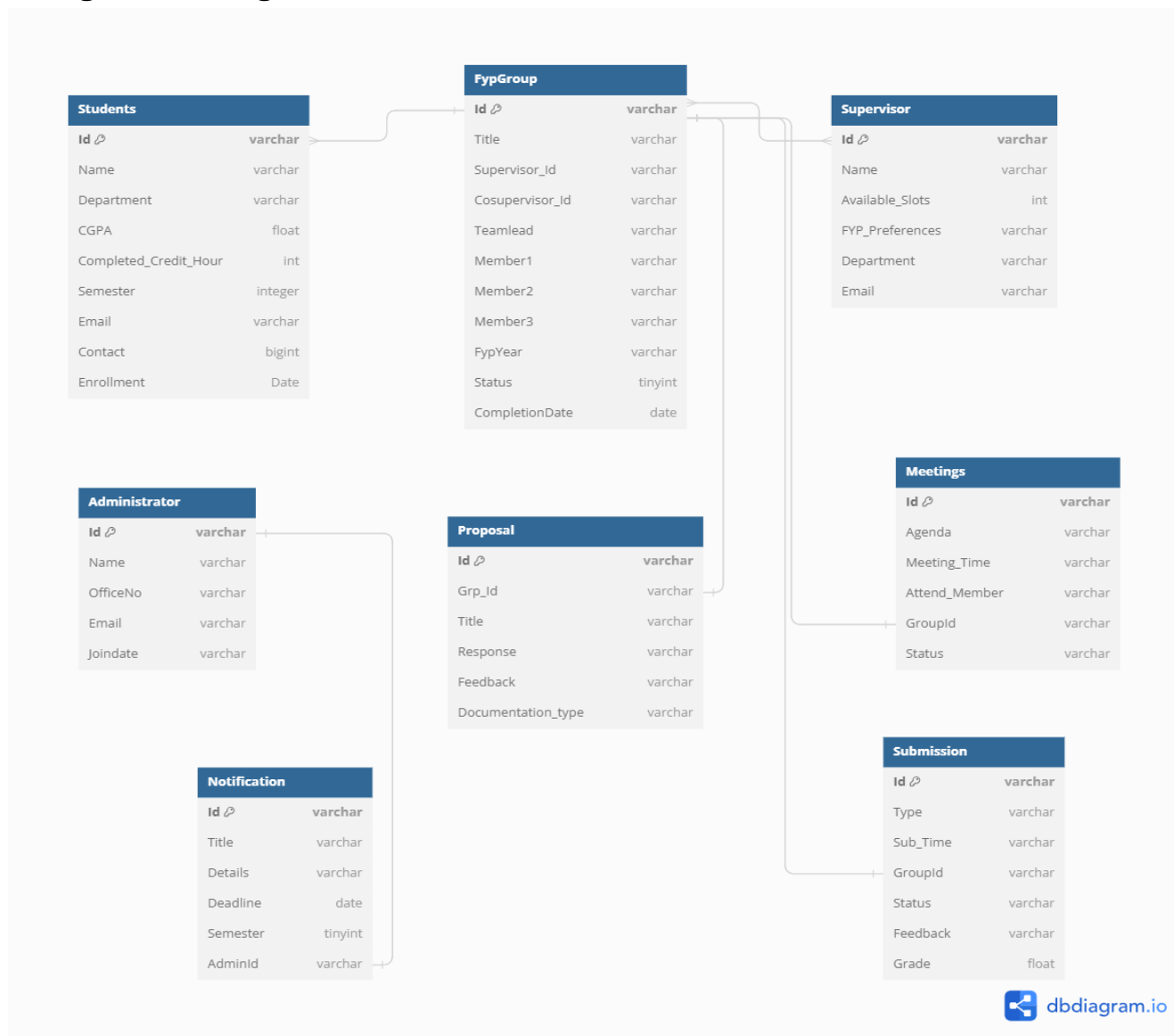


Figure 17: Design Class Diagram

5.2 Database Design

5.2.1 ER Diagram

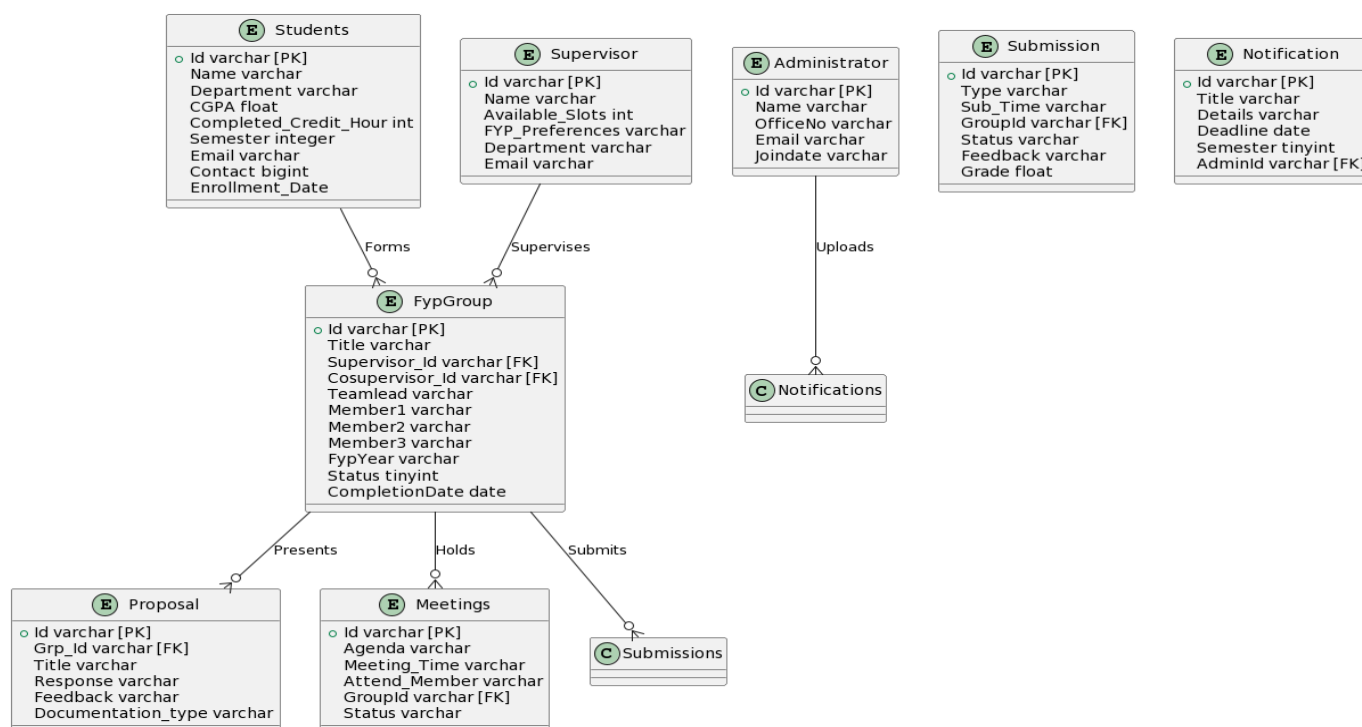


Figure 18: ER Diagram

5.2.2 Data Dictionary

5.2.2.1 Data 1

Data 1						
Name	Students					
Alias	std					
Where-used/how-used	Primary entity to store student data					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
Name	student name	Varchar	100 bytes	No	""	
ID	student id	UUID	50 bytes	No	""	PK
Password	account password	Varchar	30 bytes	Yes	""	
Credit hours	Completed credit hours	int	10 bytes	No	""	
CGPA	Student grade	float	10 bytes	No	2.00	
Semester	current semester	int	10 bytes	No	7	
Email	student email	Varchar	50 bytes	No	""	
Enrollment	Enrollment date	Datetime	50 bytes	No	""	

5.2.2.2 Data 2

Data 2						
Name		Supervisors				
Alias		spv				
Where-used/how-used		Primary entity to store supervisor data				
Content description		Notation for representing content.				
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Name</i>	<i>supervisor name</i>	<i>Varchar</i>	<i>100 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>ID</i>	<i>supervisor id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Password</i>	<i>account password</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>“ ”</i>	
<i>Preferences</i>	<i>area of interest</i>	<i>Varchar</i>	<i>250</i>	<i>No</i>	<i>“ ”</i>	
<i>Slots</i>	<i>Available slots</i>	<i>Varchar</i>	<i>100</i>	<i>No</i>	<i>“ ”</i>	
<i>Email</i>	<i>email</i>	<i>Varchar</i>	<i>50</i>	<i>No</i>	<i>“ ”</i>	

5.2.2.3 Data 3

Data 3						
Name		FYP				
Alias		fyp				
Where-used/how-used		Primary entity to store current fyp data				
Content description		Notation for representing content.				
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Title</i>	<i>Project name</i>	<i>Varchar</i>	<i>100 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>ID</i>	<i>Project id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Supervisor Id</i>	<i>supervisor of the project</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>FK</i>
<i>Co Supervisor Id</i>	<i>Completed credit hours</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>Yes</i>	<i>“ ”</i>	<i>FK</i>
<i>Student 1</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>FK</i>

<i>Student 2</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>““</i>	<i>FK</i>
<i>Student 3</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>yes</i>	<i>““</i>	<i>FK</i>
<i>Student 4</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>Yes</i>	<i>““</i>	<i>FK</i>
<i>Year</i>	<i>Year of completion</i>	<i>int</i>	<i>20 bytes</i>	<i>Yes</i>	<i>“”</i>	
<i>Status</i>	<i>Is the project completed?</i>	<i>Tinyint</i>	<i>1 byte</i>	<i>No</i>	<i>“”</i>	

5.2.2.4 Data 4

Data 4						
Name	Submissions					
Alias	subs					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>ID</i>	<i>document id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“”</i>	<i>PK</i>
<i>Project Id</i>	<i>project id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“”</i>	<i>FK</i>
<i>Type</i>	<i>Type (image,file etc)</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>“”</i>	
<i>Feedback</i>	<i>Feedback on the submission</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>Yes</i>	<i>“”</i>	
<i>Grade</i>	<i>grade</i>	<i>Float</i>	<i>4 bytes</i>	<i>Yes</i>	<i>“”</i>	

5.2.2.5 Data 5

Data 5						
Name	Submissions					
Alias	subs					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>ID</i>	<i>document id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Project Id</i>	<i>project id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>FK</i>
<i>Type</i>	<i>Type (image,file etc)</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Feedback</i>	<i>Feedback on the submission</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>Yes</i>	<i>“ ”</i>	
<i>Grade</i>	<i>grade</i>	<i>Float</i>	<i>4 bytes</i>	<i>Yes</i>	<i>“ ”</i>	

5.2.2.6 Data 6

Data 6						
Name	Admin					
Alias	admin					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>ID</i>	<i>notification id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Name</i>	<i>Admin name</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Email</i>	<i>Admin email</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Designation</i>	<i>designation inside the institution</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>“ ”</i>	

5.2.2.7 Data 7

Data 7						
Name	Meetings					
Alias	mtn					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>ID</i>	<i>meeting id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Agenda</i>	<i>purpose of the meeting</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Group Id</i>	<i>group organizing the meeting</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Status</i>	<i>Ongoing or finished?</i>	<i>tinyint</i>	<i>1 bytes</i>	<i>No</i>	<i>“ ”</i>	

5.2.2.8 Data 7

Data 8						
Name	Proposal					
Alias	prop					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
ID	proposal id	UUID	50 bytes	No	""	PK
Group Id	Group that owns the proposal	UUID	50 bytes	No	""	FK
Feedback	feedback on the proposal	Varchar	200 bytes	No	""	FK
Designation	designation inside the institution	Varchar	30 bytes	Yes	""	

5.3 Application Design

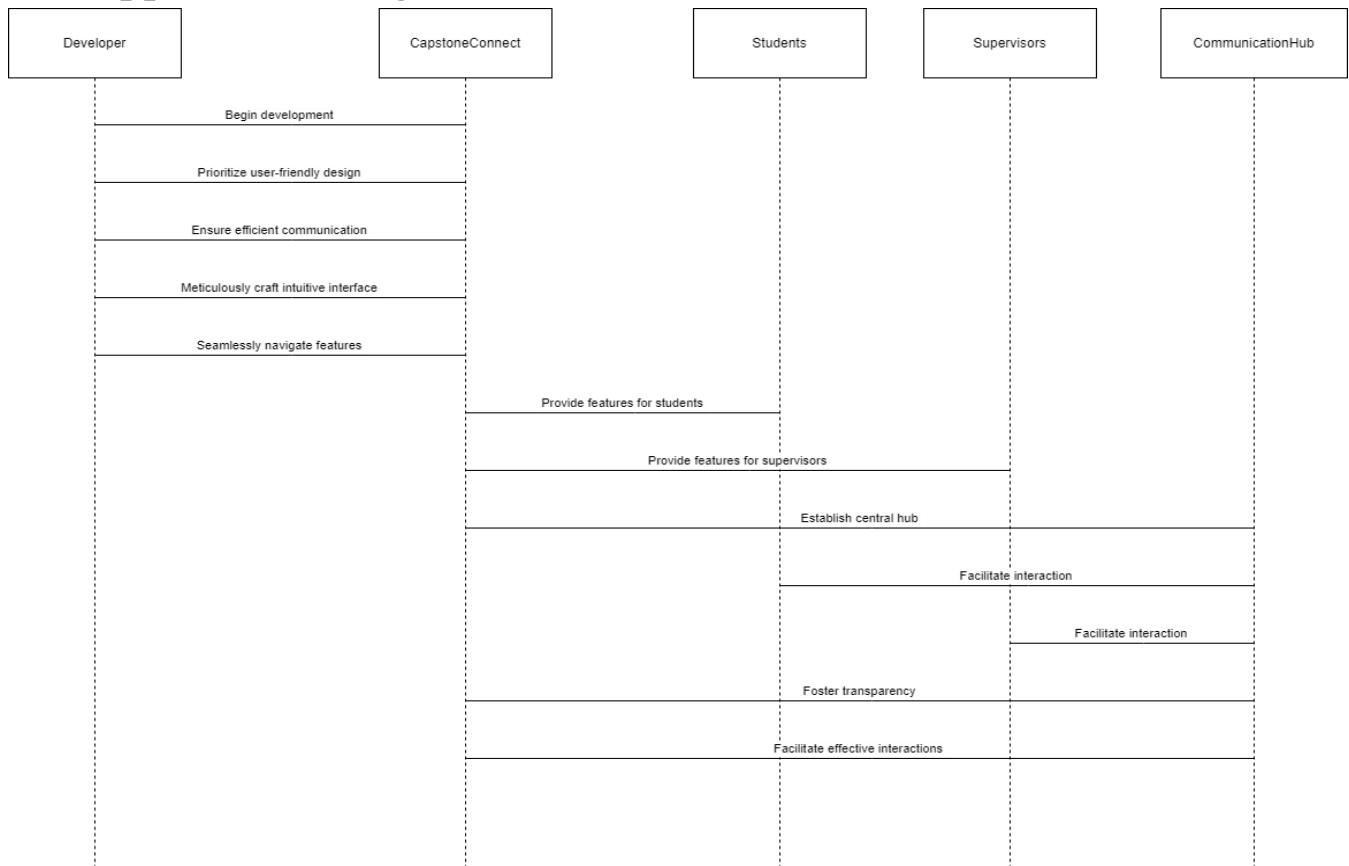


Figure 19: Application Design

5.3.1 Sequence Diagram

5.3.1.1 <Sequence Diagram - Student>

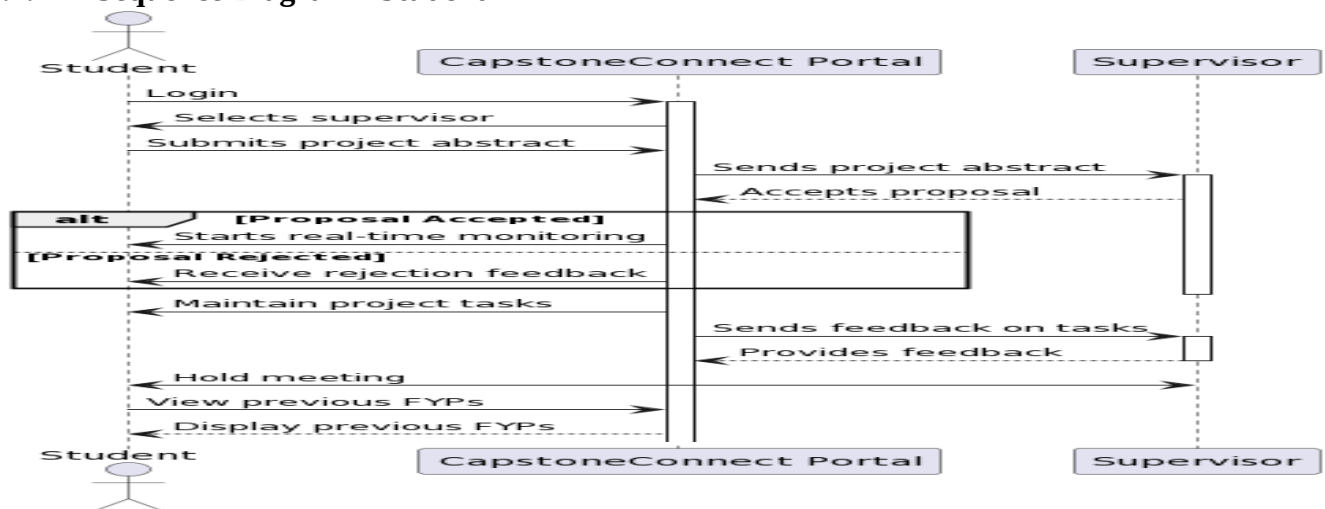


Figure 20: Sequence Diagram: Student

The student interface of CapstoneConnect provides a user-friendly platform for students to seamlessly interact with their supervisors throughout the duration of their final year project (FYP). Upon logging in, students can choose their preferred supervisor, submit project abstracts, and initiate real-time monitoring upon acceptance of their proposal by the supervisor. In case of rejection, supervisors provide constructive feedback to guide students. Throughout the project lifecycle, students can efficiently manage tasks, receive feedback on their progress, schedule meetings with supervisors, and access a repository of previous FYPs for reference, all within the intuitive interface of CapstoneConnect, fostering effective communication and collaboration between students and supervisors.

5.3.1.2 <Sequence Diagram - Supervisor>

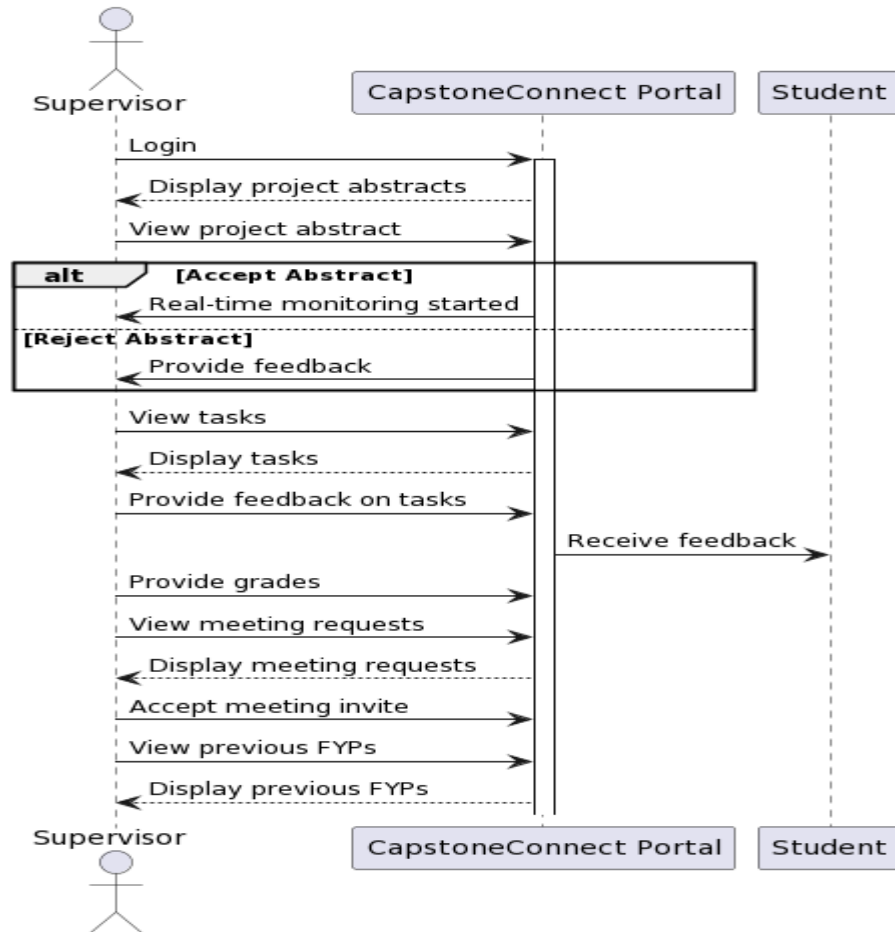


Figure 21: Sequence Diagram: Supervisor

In the CapstoneConnect portal, supervisors begin by logging in, whereupon they can access project abstracts submitted by students. If a supervisor accepts a project abstract, real-time monitoring commences, enabling supervision of the FYP group; otherwise, feedback is provided to guide the students. Supervisors can then view tasks submitted by students, offer feedback, and assign grades on completed tasks. Additionally, they can manage meeting requests, accepting invitations as per their availability. Furthermore, supervisors have the capability to review previous batch FYPs, providing valuable insights and context for their current supervisory responsibilities. This seamless interface facilitates efficient communication and collaboration between supervisors and students throughout the FYP process.

5.3.1.3 <Sequence Diagram - Administrator>

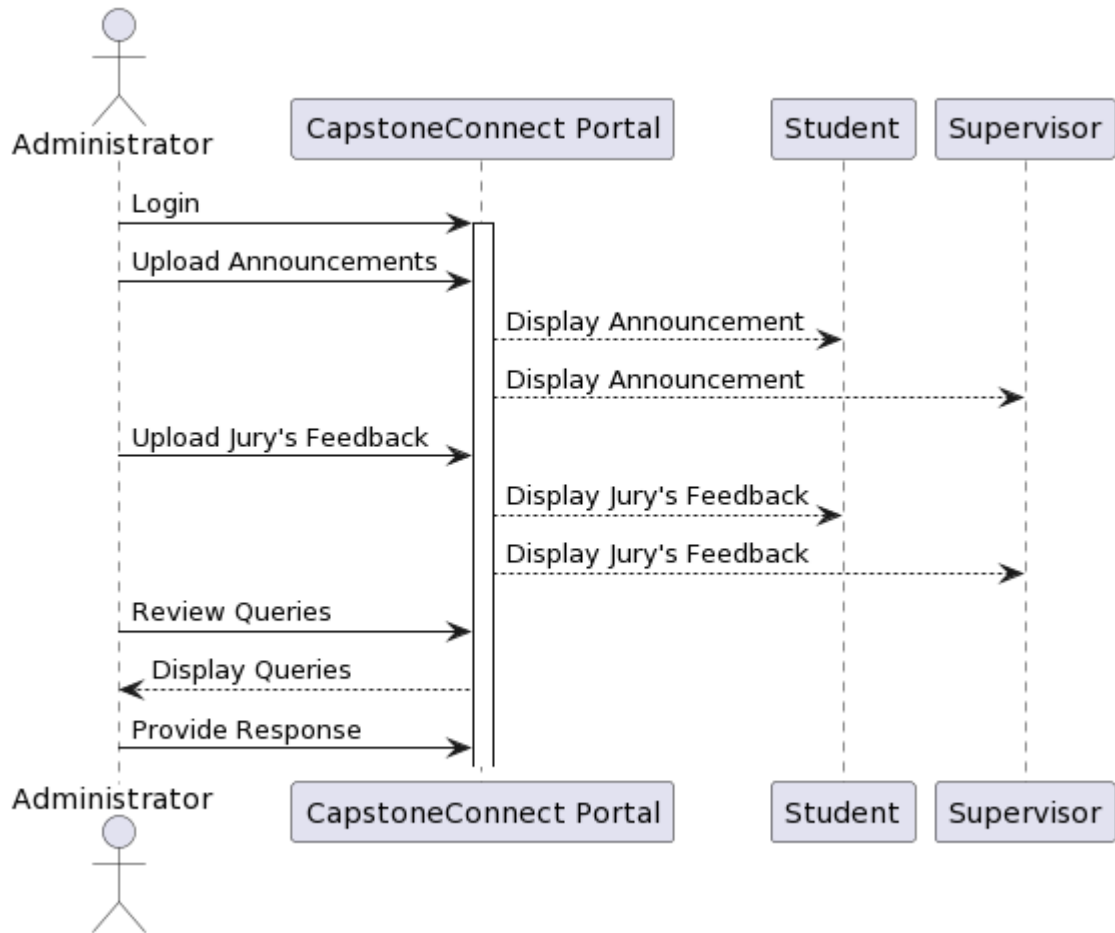


Figure 22: Sequence Diagram Administrator

In the CapstoneConnect portal, the Administrator initiates by logging in, gaining access to crucial functionalities. They upload announcements relevant to the FYP process, ensuring they are promptly displayed to both students and supervisors. Additionally, the Administrator uploads feedback documents from jury members regarding presentations and proposal defenses, which are equally accessible across all interfaces. Furthermore, they review queries submitted by both students and supervisors in real-time, providing timely responses to ensure efficient resolution of concerns and facilitate seamless communication within the CapstoneConnect platform. This interface plays a pivotal role in maintaining transparency, facilitating effective communication, and fostering a conducive environment for the successful completion of FYPs.

5.3.2 State Diagram

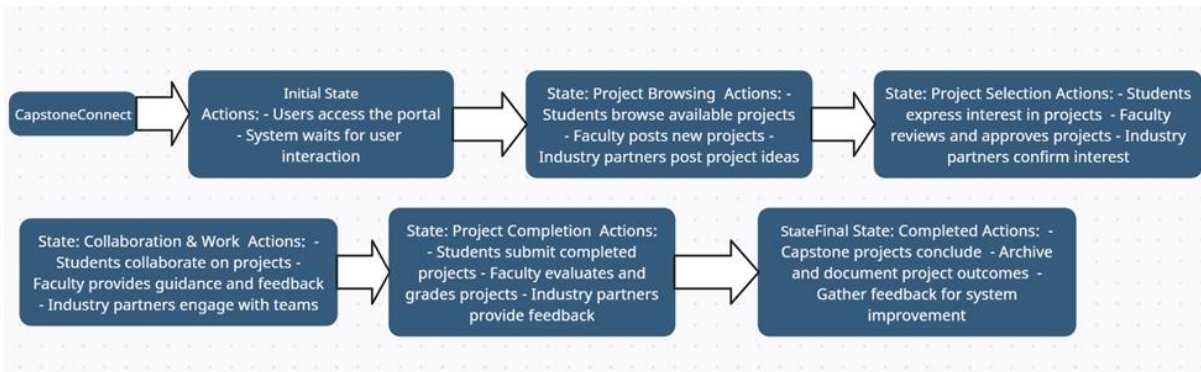


Figure 23: State Diagram

5.4 GUI Design

5.4.1 Login Module

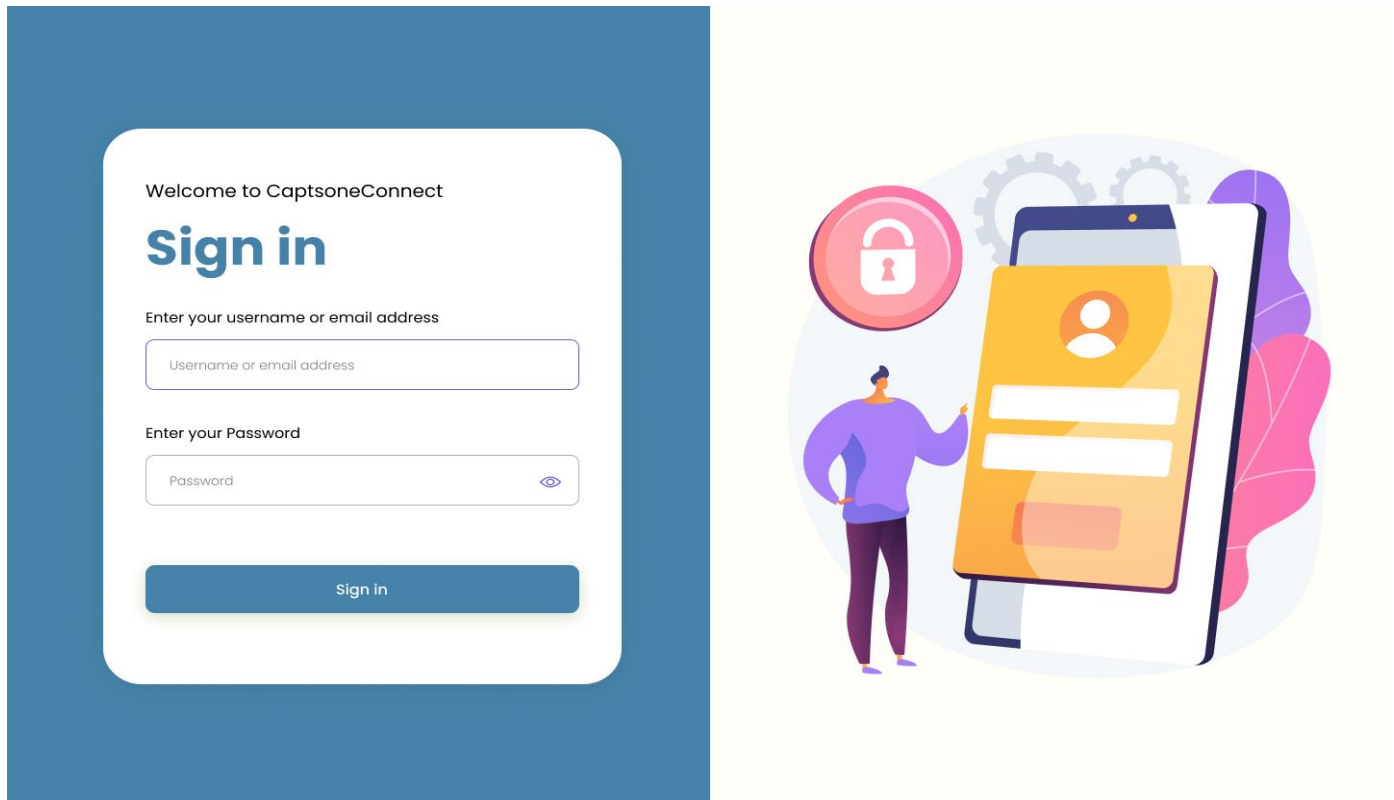


Figure 24: Login

5.4.2 Available Supervisors List

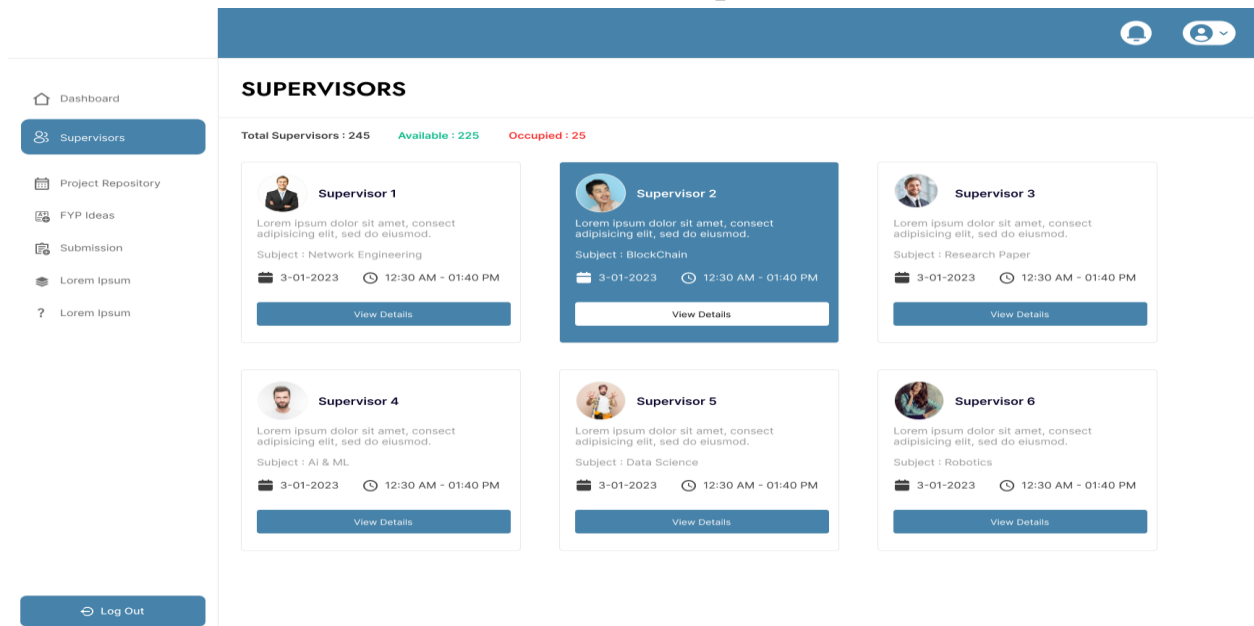


Figure 25: Available Supervisor List

5.4.3 Student Dashboard

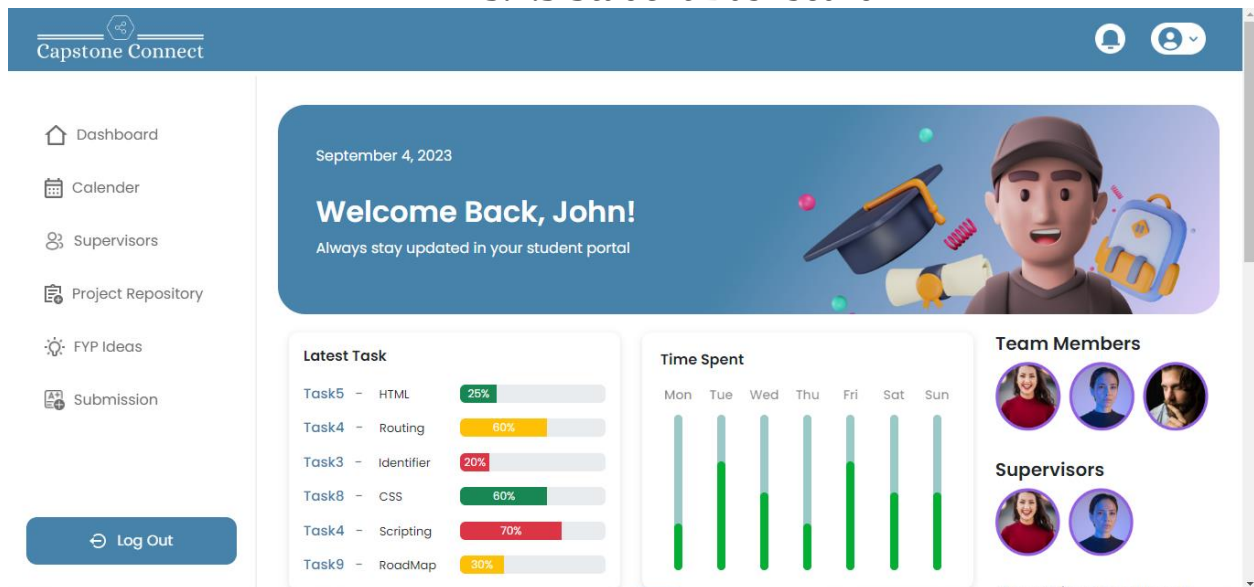


Figure 26: Student Dashboard

5.4.4 Projects list for Supervisors

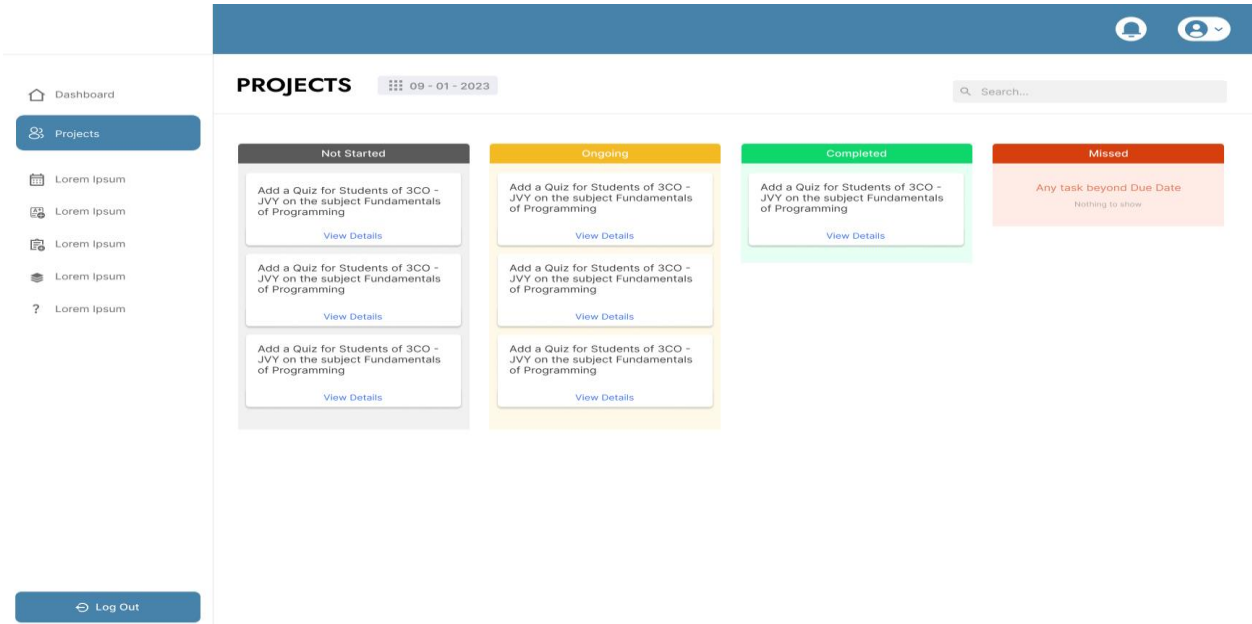


Figure 27: Projects list for Supervisors

5.4.5 FYP Calendar

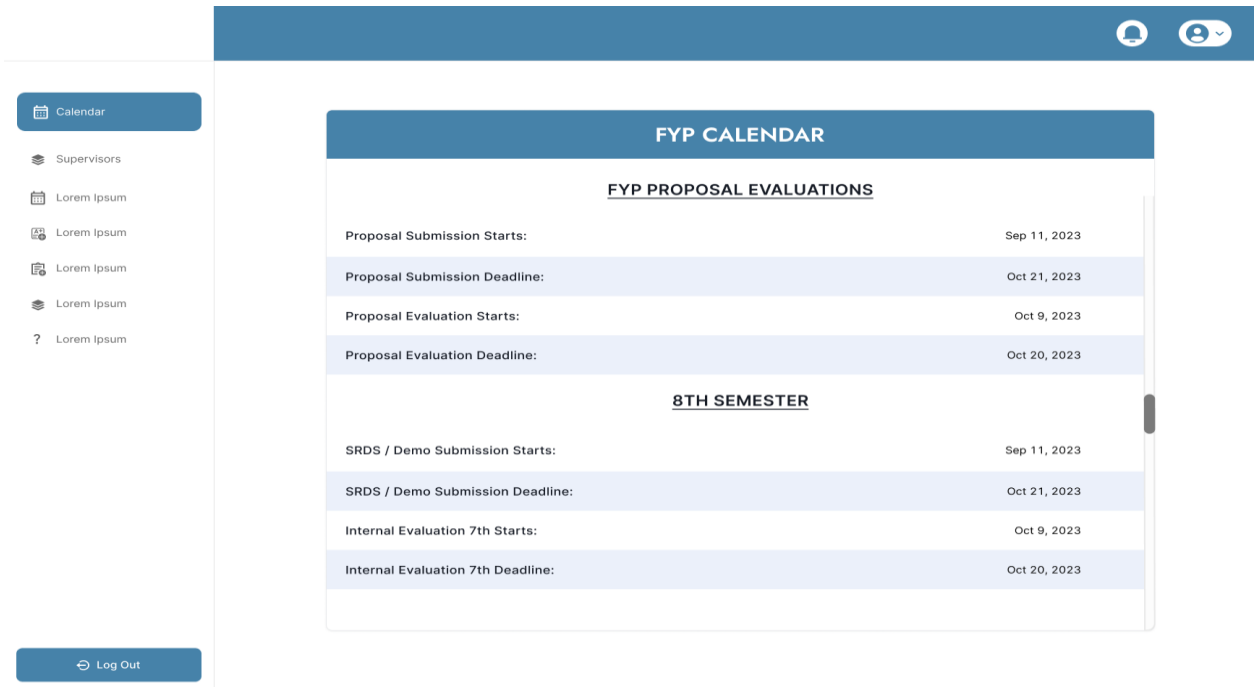
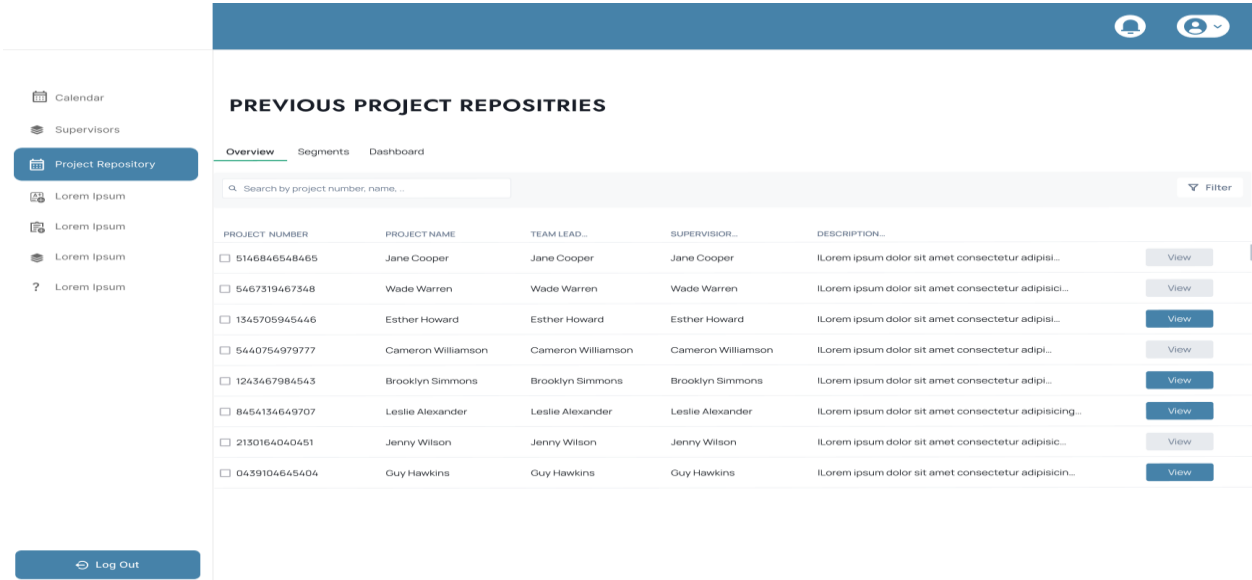


Figure 28: FYP Calendar

5.4.6 Previous Project Repository



5.4.6 Previous Project Repository

Figure 29: Previous Project Repository

5.4.7 Submission Portal

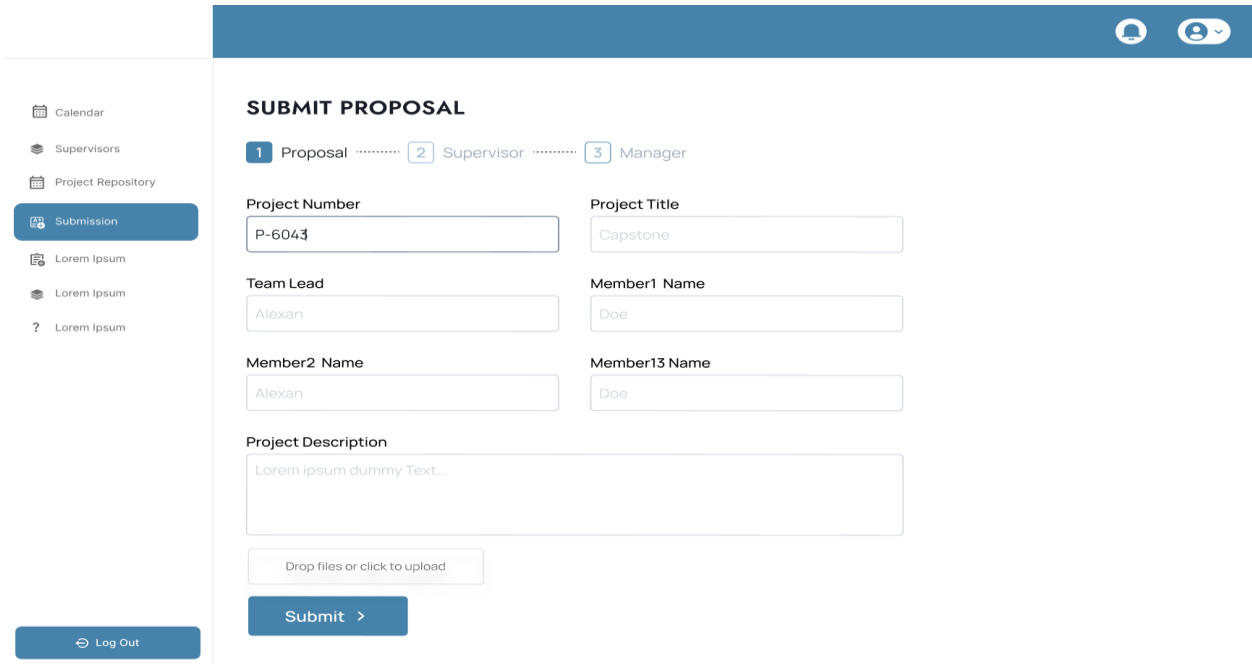


Figure 30: Submission Portal

5.4.8 Project Proposal View for Supervisor

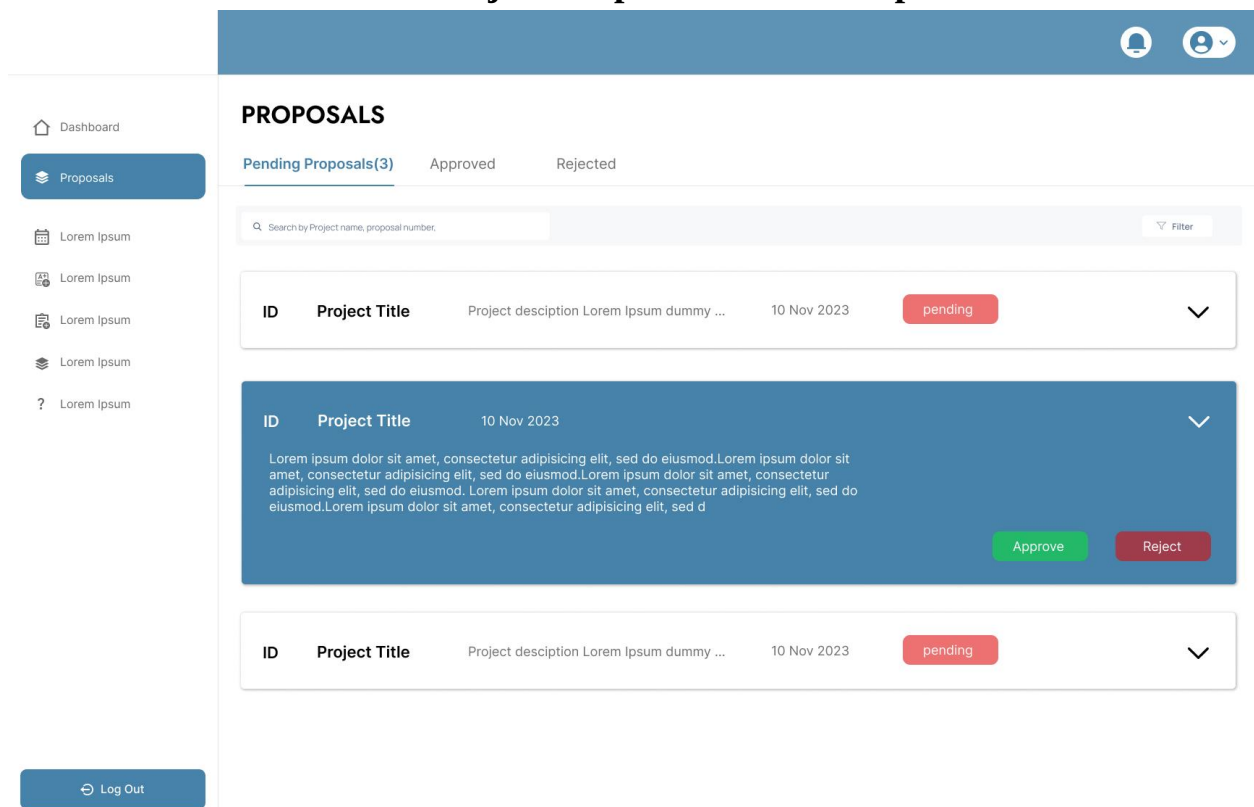


Figure 31: Project Proposal View for Supervisor

6. References

MutholibK, Abdul., Surya Gunawan, and Mira kartiwi. 2011. "Development of FYP online system for outcome based education." *TENCON IEEE*, 1312-1316.

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Azemi, and Mohamad faez. n.d. "Web Portal: A Case Study FOR UTP CENTRE FOR GRADUATE STUDIES." UTEPEDIA. <https://utpedia.utp.edu.my/id/eprint/13942/>.

MutholibK, Abdul., Surya Gunawan, and Mira kartiwi. 2011. "Development of FYP online system for outcome based education." *TENCON IEEE*, 1312-1316.

<https://ieeexplore.ieee.org/abstract/document/6129020>.

R, Affandi, TC Bee, and A. Baharom. n.d. *YEAR PROJECT (FYP) MANAGEMENT SYSTEM*.

A4. OTHER TECHNICAL DETAIL DOCUMENTS

Software Test Plan

S. No	Description	Test Engineer	Start Date	End Date
1	Login Screen	Ayushmann Kukreja	1-Feb-24	1-Feb-24
2	Registration	Ayushmann Kukreja	5-Feb-24	7-Feb-24
Admin Screens				
3	Registration Approval/Rejection	Murtaza Mehdi	10-Feb-24	11-Feb-24
4	Ongoing Project	Ayushmann Kukreja	30-Apr-24	3-May-24
Supervisor Screens				
5	Proposals	Murtaza Mehdi	15-Mar-24	18-Mar-24
6	Supervisor Dashboard	Ayushmann Kukreja	20-Mar-24	23-Mar-24
7	FYP Submissions View	Murtaza Mehdi	25-Mar-24	27-Mar-24
8	FYP Progress Tracking	Ayushmann Kukreja	20-Jun-24	23-Jun-24
FypGroup Screens				
9	Submission	Ayushmann Kukreja	1-Jun-24	3-Jun-24
10	FypGroup Dashboard	Murtaza Mehdi /	10-Jun-24	17-Jun-24
Screens/Features For All Interfaces				
11	Project Repository	Ayushmann Kukreja	5-May-24	6-May-24
12	Calendar	Murtaza Mehdi	7-May-24	8-May-24
13	Guidelines	Ayushmann Kukreja	9-May-24	10-May-24
14	Supervisors	Murtaza Mehdi	11-May-24	12-May-24
15	Queries	Ayushmann Kukreja	13-May-24	14-May-24
16	Notifications	Murtaza Mehdi	10-Jun-24	13-Jun-24
17	Chat / Communication Channels	Ayushmann Kukreja	15-Jun-24	20-Jun-24
18	User Profile	Murtaza Mehdi	25-Jun-24	28-Jun-24
19	Fortnightly Signup Sheet	Ayushmann Kukreja	29-Jun-24	30-Jun-24

Capstone Connect Test Cases

Test Case 1: Login Screen

Iteration No: 1

Module Name: **Login Screen**

Date: **1-Feb-24**

Test Case ID: **LOGIN_001**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Test the functionality of the login screen.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Testing with valid credentials					
TC-1	Enter User-ID	cs301245	User ID entered	User ID entered	Pass
	Enter Password	testsupervisor	Password entered	Password entered	Pass
	Press [Enter] button		Login into application	Allowed to enter into application	Pass
Testing with invalid credentials					
TC-2	Enter User-ID	cs301245	User ID entered	User ID entered	Pass
	Enter Password	testsupervisor123	Password entered	Incorrect password error	Pass
	Press [Enter] button		Display error message	Error message displayed	Pass

Test Case 2: Registration

Iteration No: 1

Module Name: **Registration**

Date: **5-Feb-24**

Test Case ID: **REG_001**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Test the functionality of the registration screen.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Testing With Correct inputs and Selecting 3 Members					
TC-1	Select No. Of Members	3	3 is selected in dropdown	3 is selected in dropdown	Pass
	Enter Project Title	Muhafiz	Project Title entered	Project Title entered	Pass
	Enter Team Lead ID	cs201632	Team Lead ID entered	Team Lead ID entered	Pass
	Get OTP on Email	OTPs on teamlead's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Valid Teamlead's OTP as provided on Email	OTP entered	OTP entered	Pass
	Enter other member's ID	Member's IDs	Member's IDs entered	Member's IDs entered	Pass

	Get OTP from members	OTPs on member's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Valid Member's OTP as provided on Email	Member's OTP entered	Member's OTP entered	Pass
	Registration Successful				Pass
Testing With Correct inputs and Selecting 4 Members					
TC-2	Select No. Of Members	4	4 is selected in dropdown	4 is selected in dropdown	Pass
	Enter Project Title	SafeDrive	Project Title entered	Project Title entered	Pass
	Enter Team Lead ID	cs201645	Team Lead ID entered	Team Lead ID entered	Pass
	Get OTP on Email	OTPs on teamlead's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Valid Teamlead's OTP as provided on Email	OTP entered	OTP entered	Pass
	Enter other member's ID	Member's IDs	Member's IDs entered	Member's IDs entered	Pass
	Get OTP from members	OTPs on member's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Valid Member's OTP as provided on Email	Member's OTP entered	Member's OTP entered	Pass
	Registration Successful				Pass
Testing With Invalid Student Id					
TC-3	Select No. Of Members	3	3 is selected in dropdown	3 is selected in dropdown	Pass
	Enter Project Title	TechFuture	Project Title entered	Project Title entered	Pass
	Enter Team Lead ID	cs2016XX	Error message for invalid ID	Error message displayed, registration halted	Pass
	Registration Failed				Pass
TC-4	Select No. Of Members	3	3 is selected in dropdown	3 is selected in dropdown	Pass
	Enter Project Title	Innovators	Project Title entered	Project Title entered	Pass
	Enter Team Lead ID	cs201623	Team Lead ID entered	Team Lead ID entered	Pass
	Get OTP on Email	OTPs on teamlead's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Invalid OTP	Error message for invalid OTP	Error message displayed, registration halted	Pass
	Registration Failed				Pass
Testing With Incorrect OTP					
TC-5	Select No. Of Members	3	3 is selected in dropdown	3 is selected in dropdown	Pass
	Enter Project Title	FutureTech	Project Title entered	Project Title entered	Pass
	Enter Team Lead ID	cs201634	Team Lead ID entered	Team Lead ID entered	Pass

	Get OTP on Email	OTPs on teamlead's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Valid Teamlead's OTP as provided on Email	OTP entered	OTP entered	Pass
	Enter other member's ID	Member's IDs	Member's IDs entered	Member's IDs entered	Pass
	Get OTP from members	OTPs on member's email	OTP Delivered	OTP Delivered	Pass
	Enter OTP	Invalid OTP	Error message for invalid OTP	Error message displayed, registration halted	Pass
	Registration Failed				Pass

Test Case 3: Registration Approval/Rejection

Iteration No: 1

Module Name: **Admin Screens**

Date: **10-Feb-24**

Test Case ID: **ADMIN_001**

Test Engineer: **Murtaza Mehdi**

Test Case Description: **Test the functionality of the registration approval/rejection process.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Admin - Registration Approval					
TC-1	Admin clicks on approve	Approve Button Clicked	Confirmation popup occurs	Confirmation popup occurs	Pass
	Admin confirms on alert box	Click on alert box	FYP group registration approved	FYP group registration approved	Pass
	FYP group's status is set to Approved				Pass
Admin - Registration Rejection					
TC-2	Admin clicks on reject	Reject Button Clicked	Confirmation popup occurs	Confirmation popup occurs	Pass
	Admin confirms on alert box	Click on alert box	FYP group registration rejected	FYP group registration rejected	Pass
	FYP group's status is set to Rejected				Pass

Test Case 4: Ongoing ProjectIteration No: **1**Module Name: **Admin Screens**Date: **30-Apr-24**Test Case ID: **ADMIN_002**Test Engineer: **Ayushmann Kukreja**Test Case Description: **Test the functionality of viewing and grading ongoing projects.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Admin Views Projects of All Status					
TC-1	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	View projects with status "Pending"	N/A	List of projects with status "Pending" displayed	List of projects with status "Pending" displayed	Pass
	View projects with status "Abstract"	N/A	List of projects with status "Abstract" displayed	List of projects with status "Abstract" displayed	Pass
	View projects with status "FYP1"	N/A	List of projects with status "FYP1" displayed	List of projects with status "FYP1" displayed	Pass
	View projects with status "FYP2"	N/A	List of projects with status "FYP2" displayed	List of projects with status "FYP2" displayed	Pass
	View projects with status "Complete"	N/A	List of projects with status "Complete" displayed	List of projects with status "Complete" displayed	Pass
Provide Proposal Defense Remarks for Abstract Status					
TC-2	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	Select project with status "Abstract"	Project ID	Project details displayed	Project details displayed	Pass
	Open Proposal Defense popup	Click on Proposal Defense button	Proposal Defense popup displayed	Proposal Defense popup displayed	Pass
	Enter Proposal Defense details	Date, Jury, Jury Remarks, Response	Proposal Defense details entered	Proposal Defense details entered	Pass
	Submit Proposal Defense remarks	Click Submit	Project status updated to "FYP1" if approved, else "Reevaluate"	Project status updated accordingly	Pass
Provide Mid Evaluation Remarks for FYP1 Status					

TC-3	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	Select project with status "FYP1"	Project ID	Project details displayed	Project details displayed	Pass
	Open Mid Evaluation popup	Click on Mid Evaluation button	Mid Evaluation popup displayed	Mid Evaluation popup displayed	Pass
	Enter Mid Evaluation details	Grades, Remarks	Mid Evaluation details entered	Mid Evaluation details entered	Pass
	Submit Mid Evaluation remarks	Click Submit	Project status updated to "FYP2"	Project status updated to "FYP2"	Pass
Provide Final Evaluation Remarks for Complete Status					
TC-4	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	Select project with status "Complete"	Project ID	Project details displayed	Project details displayed	Pass
	Open Final Evaluation popup	Click on Final Evaluation button	Final Evaluation popup displayed	Final Evaluation popup displayed	Pass
	Enter Final Evaluation details	Grades, Remarks	Final Evaluation details entered	Final Evaluation details entered	Pass
	Submit Final Evaluation remarks	Click Submit	Project status updated to "Done" and grades emailed to team lead	Project status updated and grades emailed	Pass
Invalid Input - Missing Grades in Mid Evaluation					
TC-5	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	Select project with status "FYP1"	Project ID	Project details displayed	Project details displayed	Pass
	Open Mid Evaluation popup	Click on Mid Evaluation button	Mid Evaluation popup displayed	Mid Evaluation popup displayed	Pass
	Enter incomplete Mid Evaluation details	Grades for only 1 out of 3 members	Error message displayed indicating missing grades	Error message displayed	Pass
Invalid Input - Missing Grades in Final Evaluation					
TC-6	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass

	Select project with status "Complete"	Project ID	Project details displayed	Project details displayed	Pass
	Open Final Evaluation popup	Click on Final Evaluation button	Final Evaluation popup displayed	Final Evaluation popup displayed	Pass
	Enter incomplete Final Evaluation details	Grades for only 1 out of 3 members	Error message displayed indicating missing grades	Error message displayed	Pass
Invalid Input - Missing Jury in Proposal Defense					
TC-7	Login as Admin	Admin credentials	Admin dashboard displayed	Admin dashboard displayed	Pass
	Select project with status "Abstract"	Project ID	Project details displayed	Project details displayed	Pass
	Open Proposal Defense popup	Click on Proposal Defense button	Proposal Defense popup displayed	Proposal Defense popup displayed	Pass
	Enter Proposal Defense details without selecting Jury	Date, Jury not selected, Jury Remarks, Response	Error message displayed indicating missing Jury selection	Error message displayed	Pass

Test Case 5: Proposal

Iteration No: 1

Module Name: **Supervisor Screens**Date: **15-March-24**Test Case ID: **SUP_001**Test Engineer: **Murtaza Mehdi**Test Case Description: **Test the functionality of Proposal Screen.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Test Case for Rejection					
TC-1	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login into application	Logged in	Pass
	Navigate to proposals screen		Proposals screen is displayed	Proposals screen displayed	Pass
	Select proposal for rejection	Proposal ID	Proposal details are displayed	Proposal details displayed	Pass
	Press [Reject] button		Proposal is marked as rejected and stored	Proposal marked as rejected and stored	Pass

	Verify proposal in rejected list	Proposal ID	Proposal is present in rejected category	Proposal in rejected category	Pass
Test Case for Approval					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login into application	Logged in	Pass
	Navigate to proposals screen		Proposals screen is displayed	Proposals screen displayed	Pass
	Select proposal for approval	Proposal ID	Proposal details are displayed	Proposal details displayed	Pass
	Press [Approve] button		Proposal is marked as approved	Proposal marked as approved	Pass
	Verify proposal in approved list	Proposal ID	Proposal is present in approved category	Proposal in approved category	Pass
	Co-supervisor approves proposal		Proposal approved by co-supervisor	Proposal approved by co-supervisor	Pass
	Verify ongoing project visibility	Proposal ID	Proposal is visible on supervisor's dashboard	Proposal visible on supervisor's dashboard	Pass

Test Case 6: Supervisor Dashboard Test Case

Iteration No: **1**

Module Name: **Supervisor Dashboard**

Date: **20-Mar-24**

Test Case ID: **SUP_002**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **To test the functionality and display of the supervisor's dashboard.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
TC-1	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login into application	Logged in	Pass
	Dashboard Should be displayed		Dashboard screen is displayed	Dashboard screen is displayed	Pass
	Projects being supervised are shown on dashboard		Supervised projects are being shown	Supervised projects are being shown	Pass
	Supervised Project's Progress is being displayed		Supervised Project's Progress is being displayed	Supervised Project's Progress is being displayed	Pass

Test Case 7: Submissions Screen Test CasesIteration No: **1**Module Name: **Submissions**Date: **25-Mar-24**Test Case ID: **SUP_003**Test Engineer: **Murtaza Mehdi**

Test Case Description: **To test the functionality and display of the submissions screen for the supervisor.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass /Fail
Functionality when a submission is set to Approved					
TC-1	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Logged into application	Logged in	Pass
	Dashboard should be displayed		Dashboard screen is displayed	Dashboard screen is displayed	Pass
	Supervisor clicks on Submission button	Click on Submission button	Submissions are displayed	Submissions are displayed	Pass
	Supervisor approves a submission	Provides feedback and selects "Approved" in dropdown	Submission is approved and progress is updated	Submission is approved and progress is updated	Pass
	Submission is approved				Pass
Functionality when a submission is set to review					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Logged into application	Logged in	Pass
	Dashboard should be displayed		Dashboard screen is displayed	Dashboard screen is displayed	Pass
	Supervisor clicks on Submission button	Click on Submission button	Submissions are displayed	Submissions are displayed	Pass
	Supervisor sets a submission to review again	Provides feedback and selects "Review Again" in dropdown	Submission is set to review again and progress is not updated	Submission is set to review again and progress is not updated	Pass

Functionality with invalid inputs.					
TC-3	Supervisor logs in	Invalid User-ID	Error message displayed	Error message displayed	Pass
		Valid Password			
	Press [Enter] button		Login fails	Login fails	Pass
		Invalid Password	Error message displayed	Error message displayed	Pass
	Press [Enter] button		Login fails	Login fails	Pass
	Supervisor clicks on Submission button	Click on Submission button	Submissions are displayed	Submissions are displayed	Pass
	Supervisor enters invalid feedback	Invalid feedback text	Error message displayed for invalid input	Error message displayed for invalid input	Pass
	Supervisor tries to approve with invalid feedback	Clicks approve with invalid feedback	Submission is not approved, error message displayed	Submission is not approved, error message displayed	Pass

Test Case 8: FYP Progress Tracking

Iteration No: 1

Module Name: **FYP Progress Tracking**

Date: **20-Jun-24**

Test Case ID: **SUP_004**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that the supervisor can view, provide feedback on current tasks of the projects they are supervising.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Provide Feedback on Tasks					
TC-1	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to dashboard		Dashboard is displayed	Dashboard displayed	Pass
	Select a project	Project ID	Project details are displayed	Project details displayed	Pass
	Navigate to FYP Progress Tracking screen		FYP Progress Tracking screen is displayed	FYP Progress Tracking screen displayed	Pass
	Select a task	Task ID	Task details are displayed	Task details displayed	Pass
	Provide feedback on the selected task	Feedback	Feedback is recorded and saved	Feedback recorded and saved	Pass

Move Task to Incomplete					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to dashboard		Dashboard is displayed	Dashboard displayed	Pass
	Select a project	Project ID	Project details are displayed	Project details displayed	Pass
	Navigate to FYP Progress Tracking screen		FYP Progress Tracking screen is displayed	FYP Progress Tracking screen displayed	Pass
	Select a task	Task ID	Task details are displayed	Task details displayed	Pass
	Move the task to incomplete	Task ID	Task status is updated to incomplete	Task status updated to incomplete	Pass
Provide Invalid Input					
S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
TC-3	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to dashboard		Dashboard is displayed	Dashboard displayed	Pass
	Select a project	Invalid Project ID	Error message indicating invalid project ID	Error message displayed	Pass
	Navigate to FYP Progress Tracking screen		System should not navigate to progress tracking	Remains on current screen	Pass
	Provide feedback on a task	Invalid Feedback	Error message indicating invalid feedback format	Error message displayed	Pass

Test Case 9: FYPGroup Submission Screen

Iteration No: 1

Module Name: **Submission**

Date: **1-Jun-24**

Test Case ID: **FYP_001**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that a student can submit a document and a notification is uploaded in the notification tab.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Submit Document					
TC-1	Navigate to submission screen		Submission screen is displayed	Submission screen displayed	Pass

	Provide user ID	Valid User-ID	User ID entered	User ID entered	Pass
	Provide password	Valid Password	Password entered	Password entered	Pass
	Select type of document	Document Type (e.g. PDF)	Document type selected	Document type selected	Pass
	Select file from local storage	File Path	File selected	File selected	Pass
	Click submit		Document submitted and notification uploaded	Document submitted and notification uploaded	Pass
Invalid User ID					
TC-2	Navigate to submission screen		Submission screen is displayed	Submission screen displayed	Pass
	Provide user ID	Invalid User-ID	Error message indicating invalid user ID	Error message displayed	Pass
Submit Without Selecting File					
TC-3	Navigate to submission screen		Submission screen is displayed	Submission screen displayed	Pass
	Provide user ID	Valid User-ID	User ID entered	User ID entered	Pass
	Provide password	Valid Password	Password entered	Password entered	Pass
	Click submit without selecting a file		Error message indicating file not selected	Error message displayed	Pass

Test Case 10: FYPGroup Dashboard

Iteration No: **1**

Module Name: **Dashboard**

Date: **10-Jun-24**

Test Case ID: **FYP_002**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that operations can be performed on dashboard of FYPGroup.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
View Proposal Defense and Mid Evaluation Remarks					
TC-1	Navigate to dashboard		Dashboard is displayed	Dashboard displayed	Pass
	View proposal defense remarks		Proposal defense remarks are visible	Proposal defense remarks visible	Pass
	View mid evaluation remarks		Mid evaluation remarks are visible	Mid evaluation remarks visible	Pass
View Progress, Team Members, Supervisors, and Feedback					
TC-2	View progress		Progress is visible	Progress visible	Pass

	View team members		Team members are visible	Team members visible	Pass
	View supervisors		Supervisors are visible	Supervisors visible	Pass
	View current feedback		Current feedback is visible	Current feedback visible	Pass
Add Task to Bucket					
TC-3	Select option to add task		Add task option is selected	Add task option selected	Pass
	Provide task details	Task details	Task details are entered	Task details entered	Pass
	Click save		Task is added to bucket	Task added to bucket	Pass
Move Task					
TC-4	Select a task to move	Task ID	Task is selected	Task selected	Pass
	Move task to a different bucket	Target bucket	Task is moved to the target bucket	Task moved to target bucket	Pass
Edit Task					
TC-5	Select a task to edit	Task ID	Task is selected	Task selected	Pass
	Edit task details	Updated task details	Task details are updated	Task details updated	Pass
	Click save		Task updates are saved	Task updates saved	Pass
Invalid Task Details					
TC-6	Provide invalid task details	Invalid task details	Error message indicating invalid task details	Error message displayed	Pass

Test Case 11: Project Repository

Iteration No: 1

Module Name: **Project Repository**

Date: **5-May-24**

Test Case ID: **AFS_001**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that AFS (ADMIN, FYPGROUP, and SUPERVISOR) can view previous batch's FYPs.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
View Project Repository (Admin Interface)					
TC-1	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Project Repository screen		Project Repository screen is displayed	Project Repository screen displayed	Pass
	View list of projects		List of projects is displayed	List of projects displayed	Pass

View Project Repository (FYGroup Interface)					
TC-2	FYGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Project Repository screen		Project Repository screen is displayed	Project Repository screen displayed	Pass
	View list of projects		List of projects is displayed	List of projects displayed	Pass
View Project Repository (Supervisor Interface)					
TC-3	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Project Repository screen		Project Repository screen is displayed	Project Repository screen displayed	Pass
	View list of projects		List of projects is displayed	List of projects displayed	Pass

Test Case 12: Calendar

Iteration No: 1

Module Name: **Calendar**

Date: **7-May-24**

Test Case ID: **AFS_002**

Test Engineer: **Murtaza Mehdi**

Test Case Description: **Verify that the admin can perform all CRUD operations (Create, Read, Update, and Delete) on the calendar. While other users can only view**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass /Fail
Admin - Perform CRUD Operations					
TC-1	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Calendar screen		Calendar screen is displayed	Calendar screen displayed	Pass
	Create a new calendar event	Event details	Event is created and visible in the calendar	Event created and visible	Pass
	Read/view an existing calendar event	Event ID	Event details are displayed	Event details displayed	Pass
	Update an existing calendar event	Updated event details	Event is updated and changes are visible	Event updated and changes visible	Pass
	Delete an existing calendar event	Event ID	Event is deleted and no longer visible	Event deleted and no longer visible	Pass
Supervisor - View Calendar					
	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass

TC-2		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Calendar screen		Calendar screen is displayed	Calendar screen displayed	Pass
	View calendar events		Calendar events are visible	Calendar events visible	Pass
FYPGroup - View Calendar					
TC-3	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Calendar screen		Calendar screen is displayed	Calendar screen displayed	Pass
	View calendar events		Calendar events are visible	Calendar events visible	Pass

Test Case 13: Guidelines

Iteration No: 1

Module Name: **Guidelines**

Date: **9-May-24**

Test Case ID: **AFS_003**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that the admin can perform all CRUD operations (Create, Read, Update, and Delete) on the guidelines. While other users can only view**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Admin - Perform CRUD Operations					
TC-1	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Guidelines screen		Guidelines screen is displayed	Guidelines screen displayed	Pass
	Create a new guideline	Guideline details	Guideline is created and visible in the list	Guideline created and visible	Pass
	Read/view an existing guideline	Guideline ID	Guideline details are displayed	Guideline details displayed	Pass
	Update an existing guideline	Updated guideline details	Guideline is updated and changes are visible	Guideline updated and changes visible	Pass
	Delete an existing guideline	Guideline ID	Guideline is deleted and no longer visible	Guideline deleted and no longer visible	Pass
Supervisor - View Calendar					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass

	Navigate to Guidelines screen		Guidelines screen is displayed	Guidelines screen displayed	Pass
	View guidelines		Guidelines are visible	Guidelines visible	Pass
FYPGroup - View Calendar					
TC-3	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Guidelines screen		Guidelines screen is displayed	Guidelines screen displayed	Pass
	View guidelines		Guidelines are visible	Guidelines visible	Pass

Test Case 14: Supervisors Screen

Iteration No: 1

Module Name: **Supervisor's Screen**

Date: **11-May-24**

Test Case ID: **AFS_004**

Test Engineer: **Murtaza Mehdi**

Test Case Description: **Verify that the admin can perform all CRUD operations (Create, Read, Update, Delete) on the supervisor's information and assign supervisor's rank. While other users can only view**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Admin - Perform CRUD Operations					
TC-1	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Supervisor's Screen		Supervisor's Screen is displayed	Supervisor's Screen displayed	Pass
	Create a new supervisor record	Supervisor details	Supervisor is created and visible in the list	Supervisor created and visible	Pass
	Read/view an existing supervisor record	Supervisor ID	Supervisor details are displayed	Supervisor details displayed	Pass
	Update an existing supervisor record	Updated supervisor details	Supervisor is updated and changes are visible	Supervisor updated and changes visible	Pass
	Delete an existing supervisor record	Supervisor ID	Supervisor is deleted and no longer visible	Supervisor deleted and no longer visible	Pass
	Assign rank to a supervisor	Supervisor ID, Rank	Supervisor's rank is assigned and visible	Supervisor's rank assigned and visible	Pass
Supervisor - View Calendar					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass

	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Supervisor's Screen		Supervisor's Screen is displayed	Supervisor's Screen displayed	Pass
	View supervisor's information		Supervisor's information is visible	Supervisor's information visible	Pass
FYPGroup - View Calendar					
TC-3	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Supervisor's Screen		Supervisor's Screen is displayed	Supervisor's Screen displayed	Pass
	View supervisor's information		Supervisor's information is visible	Supervisor's information visible	Pass

Test Case 15: Queries

Iteration No: 1

Module Name: **Queries**

Date: **13-May-24**

Test Case ID: **AFS_005**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that the FYP group can submit their queries and Admin can provide response to them.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
FYPGroup - Submit Queries					
TC-1	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Queries screen		Queries screen is displayed	Queries screen displayed	Pass
	Select option to submit a query		Query submission form is displayed	Query submission form displayed	Pass
	Provide query details	Query details	Query details are entered	Query details entered	Pass
	Click submit		Query is submitted successfully	Query submitted successfully	Pass
Admin - View and Answer Queries					

TC-2	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Queries screen		Queries screen is displayed	Queries screen displayed	Pass
	View list of submitted queries		List of queries is displayed	List of queries displayed	Pass
	Select a query to answer	Query ID	Query details are displayed	Query details displayed	Pass
	Provide answer to the query	Answer details	Answer is recorded and visible to the FYP group	Answer recorded and visible	Pass

Test Case 16: Notification

Iteration No: **1**

Module Name: **Notification**

Date: **10-Jun-24**

Test Case ID: **AFS_006**

Test Engineer: **Murtaza Mehdi**

Test Case Description: **Verify that a notification is generated when AFS performs an event.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
Admin - Generate Notification on Event					
TC-1	Admin logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Perform an event (e.g., create a guideline)	Event details	Event is performed successfully	Event performed successfully	Pass
	Navigate to Notification tab		Notification is displayed	Notification displayed	Pass
Supervisor - Generate Notification on Event					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Perform an event (e.g., approve a proposal)	Event details	Event is performed successfully	Event performed successfully	Pass
	Navigate to Notification tab		Notification is displayed	Notification displayed	Pass

FYPGroup - Generate Notification on Event					
TC-3	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Perform an event (e.g., submit a document)	Event details	Event is performed successfully	Event performed successfully	Pass
	Navigate to Notification tab		Notification is displayed	Notification displayed	Pass

Test Case 17: Chat / Communication Channels

Iteration No: 1

Module Name: **Chat / Communication Channels**

Date: **15-Jun-24**

Test Case ID: **AFS_007**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that the FYP group can send messages to their supervisor and co-supervisor and they can respond back.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
FYPGroup - Send Message to Supervisor/Co-Supervisor					
TC-1	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Chat/Communication Channels screen		Chat/Communication Channels screen is displayed	Chat/Communication Channels screen displayed	Pass
	Select supervisor/co-supervisor	Supervisor/Co-Supervisor ID	Supervisor/Co-Supervisor is selected	Supervisor/Co-Supervisor selected	Pass
	Type message	Message content	Message is typed	Message typed	Pass
	Click send		Message is sent successfully	Message sent successfully	Pass
Supervisor - Reply to FYPGroup					
TC-2	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass

	Navigate to Chat/Communication Channels screen		Chat/Communication Channels screen is displayed	Chat/Communication Channels screen displayed	Pass
	Select FYP group	FYP group ID	FYP group is selected	FYP group selected	Pass
	View message		Message is displayed	Message displayed	Pass
	Type reply	Reply content	Reply is typed	Reply typed	Pass
	Click send		Reply is sent successfully	Reply sent successfully	Pass

Test Case 18: User Profile

Iteration No: 1

Module Name: **User Profile**

Date: **25-Jun-24**

Test Case ID: **AFS_008**

Test Engineer: **Murtaza Mehdi**

Test Case Description: **Verify that all users (admin, supervisor, FYP group) can view their profile details and change passwords**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
View User Profile Details					
TC-1	User logs in (Admin, Supervisor, FYPGroup)	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to User Profile screen		User Profile screen is displayed	User Profile screen displayed	Pass
	View profile details		Profile details are visible	Profile details visible	Pass
Change Password					
TC-2	User logs in (Admin, Supervisor, FYPGroup)	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to User Profile screen		User Profile screen is displayed	User Profile screen displayed	Pass
	Select option to change password		Change password form is displayed	Change password form displayed	Pass
	Enter current password	Current Password	Current password is entered	Current password entered	Pass
	Enter new password	New Password	New password is entered	New password entered	Pass
	Confirm new password	Confirm New Password	New password is confirmed	New password confirmed	Pass

	Click save		Password is changed successfully	Password changed successfully	Pass
Invalid Current Password					
TC-3	User logs in (Admin, Supervisor, FYPGroup)	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to User Profile screen		User Profile screen is displayed	User Profile screen displayed	Pass
	Select option to change password		Change password form is displayed	Change password form displayed	Pass
	Enter invalid current password	Invalid Current Password	Error message indicating invalid current password	Error message displayed	Pass

Test Case 19: Fortnightly Signup Sheet

Iteration No: 1

Module Name: **Fortnightly Signup Sheet**

Date: **29-Jun-24**

Test Case ID: **AFS_009**

Test Engineer: **Ayushmann Kukreja**

Test Case Description: **Verify that the FYP group can perform CRUD operations (Create, Read, Update, and Delete) on the Minutes of Meeting (MOM) which appears on the Fortnightly Signup Sheet. While Supervisor can review them.**

S. No	Steps	Input Data	Expected Result	Actual Result	Pass/Fail
FYPGroup - Perform CRUD Operations on Minutes of Meeting					
TC-1	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Fortnightly Signup Sheet screen		Fortnightly Signup Sheet screen is displayed	Fortnightly Signup Sheet screen displayed	Pass
	Create a new MOM entry	MOM details	MOM is created and visible in the list	MOM created and visible	Pass
	Read/view an existing MOM entry	MOM ID	MOM details are displayed	MOM details displayed	Pass

	Update an existing MOM entry	Updated MOM details	MOM is updated and changes are visible	MOM updated and changes visible	Pass
	Delete an existing MOM entry	MOM ID	MOM is deleted and no longer visible	MOM deleted and no longer visible	Pass
FYPGroup - Download MOM and Fortnightly Signup Sheet as PDF					
TC-2	FYPGroup logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Fortnightly Signup Sheet screen		Fortnightly Signup Sheet screen is displayed	Fortnightly Signup Sheet screen displayed	Pass
	Select option to download MOM as PDF		MOM is downloaded as PDF	MOM downloaded as PDF	Pass
	Select option to download Fortnightly Signup Sheet as PDF		Fortnightly Signup Sheet is downloaded as PDF	Fortnightly Signup Sheet downloaded as PDF	Pass
Supervisor - View Fortnightly Signup Sheet and MOM					
TC-3	Supervisor logs in	Valid User-ID	User ID entered	User ID entered	Pass
		Valid Password	Password entered	Password entered	Pass
	Press [Enter] button		Login to application	Logged in	Pass
	Navigate to Fortnightly Signup Sheet screen		Fortnightly Signup Sheet screen is displayed	Fortnightly Signup Sheet screen displayed	Pass
	View list of MOM entries		List of MOM entries is displayed	List of MOM entries displayed	Pass
	View Fortnightly Signup Sheet		Fortnightly Signup Sheet is displayed	Fortnightly Signup Sheet displayed	Pass

UI/UX Detail Document

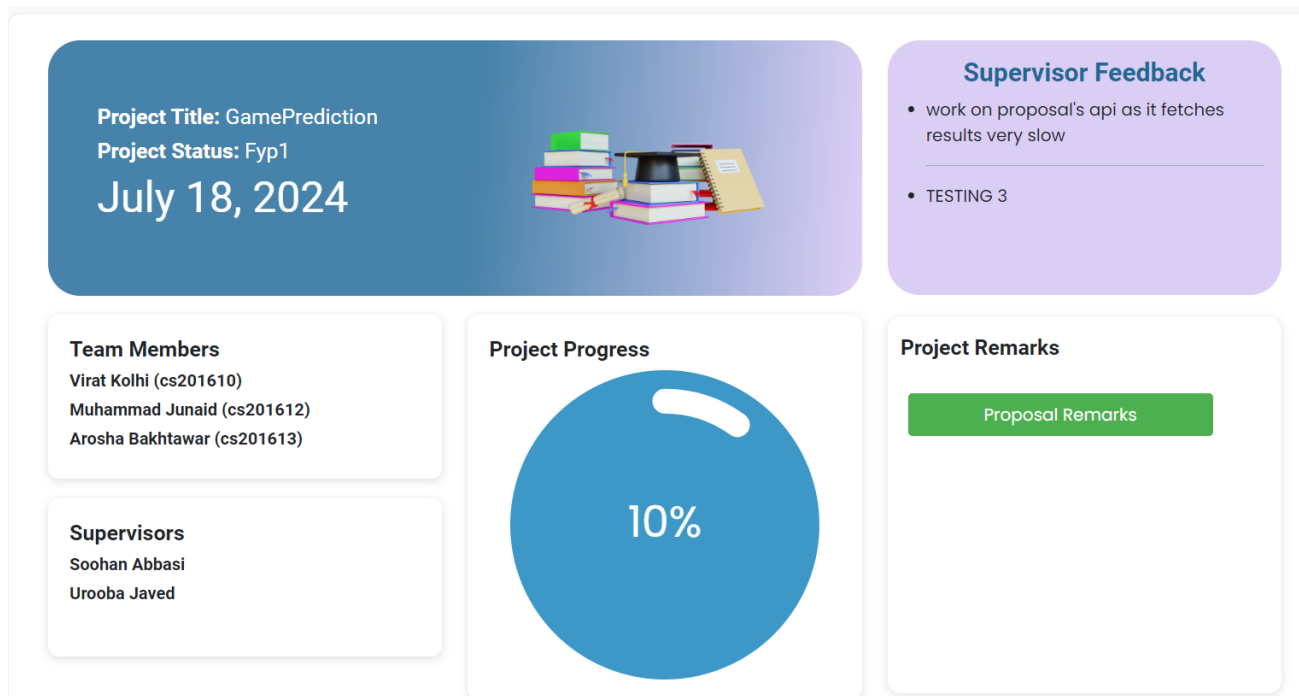


Figure 32: Student Dashboard

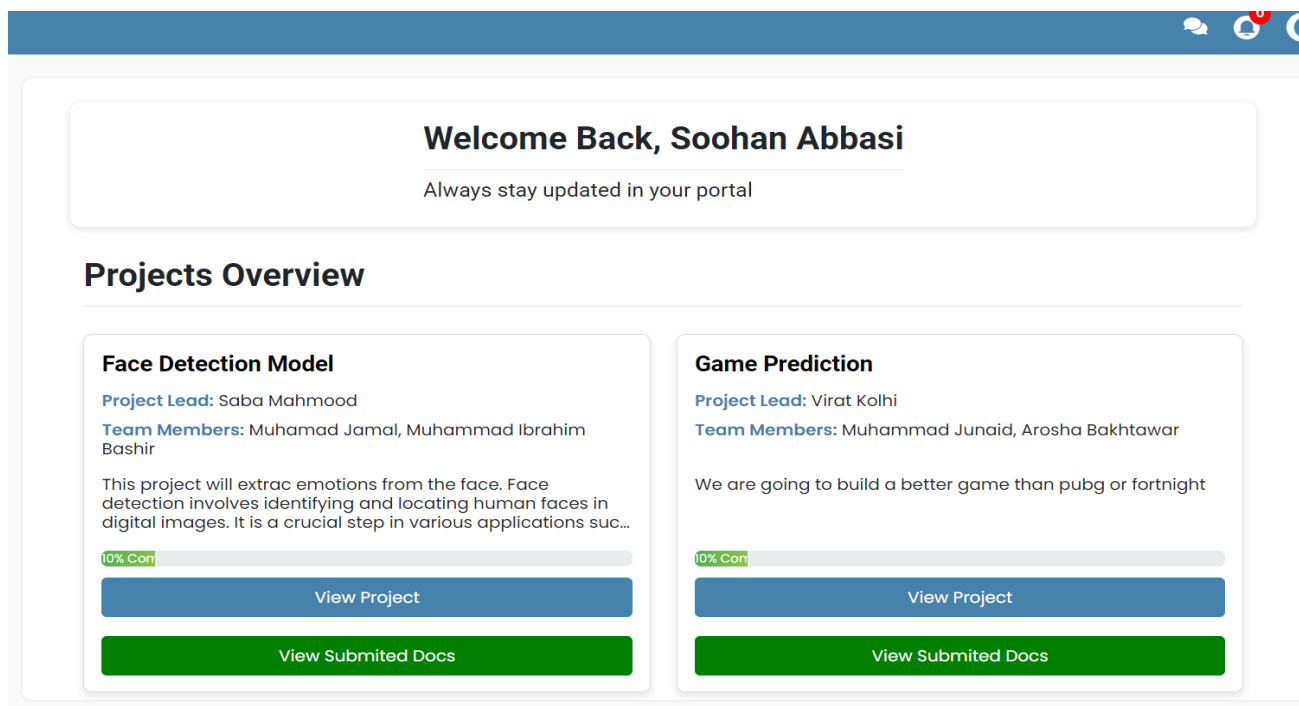


Figure 33: Supervisor Dashboard

Supervisors

Total Supervisors : 13

Dr. Najeeb Ur
Rehman Malik



ID: cs201263

Email: Najeeb@gmail.com

Department: Electronics Engineering,
Computer Science

FypPreferences: Artificial Intelligenc...

Dr. Raazia Sosan
Waseem



ID: cs201264

Email: Raazia@gmail.com

Department: Computer Science

FypPreferences: Data Science, High ...

Soohan Abbasi



ID: cs301245

Email: soohan@gmail.com

Department: Computer Science

FypPreferences: Web Development

Zahid Hussain



ID: cs301246

Email: zahid@gmail.com

Department: Computer Science

Yusra Shahina



ID: cs301247

Email: yusra@gmail.com

Department: Data Science

Yusra Mansoor



ID: cs301248

Email: yusramansoor@gmail.com

Department: Computer Science

Figure 34: Department Supervisor's Screen

Project Repository

Search By Name...

Project Number	Project Name	Team Lead	Supervisor	Co Supervisor	Action
1000	RPA	Alisha Saleem	Urooba Javed	Urooj Waheed	View
1001	Alumni Connect	Saba Mahmood	Urooj Waheed	Urooba Javed	View
2063	Health Care Centre	Babar Azam	Urooba Javed	Urooj Waheed	View
2070	CapstoneConnect	Ayushmann Kukreja	Urooba Javed	Urooj Waheed	View
2071	RPA	KIRAN SHAHZADI	Urooba Javed	Urooj Waheed	View
2074	Image Classification	Faryal nishat	Soohan Abbasi	Urooba Javed	View
2085	Image Recognition	Hamna Zahid	Urooba Javed	Urooj Waheed	View
2086	Supply Chain Management	Maryam Noreen	Sumera Hashmi	Urooj Waheed	View

1 2

Figure 35: Project Repository

Submitted Documents

Id	File Name	Project Title	Submitted By	Date	Status	Plagiarism	Feedback	Download
21	Abstract	Game Prediction	Virat Kolhi	05/21/2024	Accepted	0%	<button>Feedback</button>	<button>Download</button>
117	SRS	Game Prediction	Muhammad Junaid	07/15/2024	Accepted	0%	<button>Feedback</button>	<button>Download</button>

1

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Figure 36: Submission Screen

OnGoing Project Details

Face Detection Model

Project ID: 2072

Status: FypI

Supervisor: Soohan Abbasi

Co-Supervisor: Sumera Hashmi

Number of Members: 3

TeamLead: Saba Mahmood

Team Members: Saba Mahmood, Muhamad Jamal, Muhammad Ibrahim Bashir

Description: This project will extrac emotions from the face. Face detection involves identifying and locating human faces in digital images. It is a crucial step in various applications such as facial recognition, emotion detection, and security systems.

View Submissions

Mid Evaluation

Proposal Remarks

Back

Figure 37: Ongoing Projects Detail Screen

Coding Standards Document

1. Formatting and indentation done by Visual Studio 2022.
2. Upper Camel Case for Components
3. Lower Camel Case for variables and states.
4. Indentation unit has two spaces.
5. Use try-catch blocks to handle exceptions and errors.
6. Folder Structure is followed.

Project Policy Document

Introduction

The Bachelor of Science in Computer Science (BS-CS) degree program at DSU is a four-year degree program. Final Year Project (FYP) course spanning over two semesters (6 cr. hrs. in Total) is offered. Passing this course is a mandatory requirement for the award of the degree. The FYP course is divided into two parts: FYP-I and FYP-II. To get enrolled in FYP-II, the student must clear FYP-I. This document will brief you about the SOPs related to the FYP course.

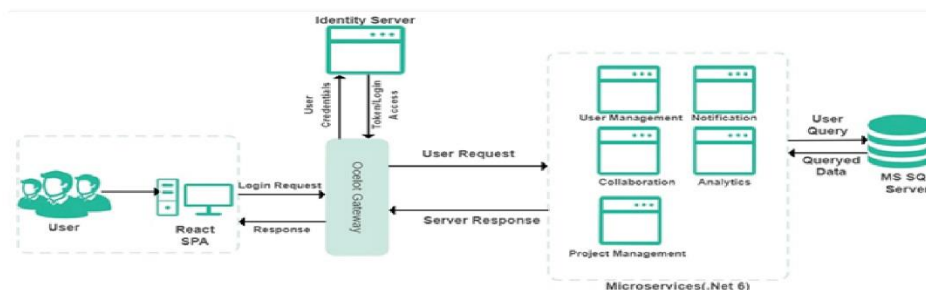
FYP Selection Method

- In the 7th semester, two-semester Final Year Project (FYP) course worth 6 credit hours is offered, provided the student has cleared 96 credit hours by the 6th semester.
- Before mid-term of the 6th semester, supervisors will submit project titles, objectives, descriptions, and expected outcomes to the PMO.
- The Reviewer committee will evaluate FYP topics and areas proposed by supervisors.
- After mid-term of the 6th semester, the list of approved FYPs and supervisors will be shared by PMO with the students.
- Students will select FYP from the approved list.
- Students may propose their FYP idea at the same time.
- The Reviewer committee will evaluate the FYP proposed by students and assign a supervisor.
- Upon selection of FYP topic, a Registration Form must be filled by the group registering for FYP-I.
- FYP proposal deadlines must be strictly followed, or individuals/group members will have to re-register for FYP-I.
- FYP proposal and report must be submitted to the PMO.
- Within 30 days, students are required to defend the proposal through a presentation in front of an evaluation panel.
- Results will be shared within 4 days by PMO.
- One chance of re-evaluation will be provided to students who fail to defend the proposal.
- On successful defense, students may start working on the FYP under the assigned supervisor.

A5. FLYER & POSTER DESIGN



APPLICATION FLOW



TOOLS



TEAM MEMBERS

SUPERVISOR – UROOJ WAHEED
CO SUPERVISOR – YUSRA MANSOOR

AYUSHMANN KUKREJA – CS201245
ABDUL REHMAN KHAN – CS201250
MURTAZA MEHDI – CS201211

A6. COPY OF EVALUATION COMMENTS

COPY OF EVALUATION COMMENTS BY SUPERVISOR FOR PROJECT – I EVALUATION

		then progress which they are lacking behind.
Urooj Waheed	P-2067 CapstoneConnect	Overall satisfactory

COPY OF EVALUATION COMMENTS BY JURY FOR PROJECT – I END SEMESTER EVALUATION

2066	R-2066 Hardware Accelerated Deep Learning for Drone-Based Garbage Detection in Pakistan	Need to broaden the scope of project
2067	P-2067 CapstoneConnect	Dataset was not available, In complete prototype.
2067	P-2067 CapstoneConnect	No doubt the project is very good but the main issue is regarding DATASET of previous FYPs which they dont have and even no sample Dataset for demonstration so they must be provided some sort of dataset to be demonstrated accordingly

**COPY OF EVALUATION COMMENTS BY SUPERVISOR FOR
PROJECT – II MID SEMESTER EVALUATION**

N/A

**COPY OF EVALUATION COMMENTS BY SUPERVISOR FOR
PROJECT – II END SEMESTER EVALUATION**

N/A

A7. MEETINGS' MINUTES

FYP-I Fortnightly Sign-Up Sheet Fall 22

Course: ☐ CS Project-1Project Code: P-2067Project Name: CapstoneConnectGroup Members Names & Reg#: Ayushmann Kukreja (CS201245) Murtaza Mehdi (CS201211) Abdul Rehman Khan (CS201250)Supervisor Name: Ms. Urooj WaheedCo-Supervisor's Name: Ms. Yusra Mansoor

Meeting #	Date	Agenda (Brief Statement)	Attended By	Supervisor's Sign	Co-supervisor's Sign	PMO's Sign
1	27/10/2024	Discussion About FYP Idea: Many innovate ideas were discussed with both supervisor	Ayushmann Mehdi Abdul Rehman			
2	30/10/2024	Finalized FYP topic and discussion about further challenges	Ayushmann Mehdi Abdul Rehman			
3	20/11/2024	Discussion about Project Proposal	Ayushmann Mehdi Abdul Rehman			
4	28/11/2024	Discussion regarding Project Defence	Ayushmann Mehdi Abdul Rehman			
5	5/12/2024	Discussion regarding conducting a survey among students and supervisors	Ayushmann Mehdi Abdul Rehman			

6	13/12/2024	Discussion about what necessary information is to be stored about student and supervisor	Ayushmann Mehdi Abdul Rehman			
7	19/12/2024	Discussion about results of survey	Ayushmann Mehdi Abdul Rehman			
8	3/1/2024	Discussion regarding making innovative user interface	Ayushmann Mehdi Abdul Rehman			
9	15/1/2024	Discussion regarding SDS and SRS	Ayushmann Mehdi Abdul Rehman			
10	21/2/2024	Further discussion about SDS (Software Architecture, Sequence Diagram)	Ayushmann Mehdi Abdul Rehman			

A8. DOCUMENT CHANGE RECORD

Date	Version	Author	Change Details
2023-06-25	1.0	Ayushmann	Initial draft creation
2023-07-10	1.1	Mehdi	Updated project scope and objectives
2023-07-18	1.2	Ayushmann	Added GUI design and database details